COUNTING THE ACTIVE DIMENSION OF A TRAINING RUN: WHAT A SINGLE SCALAR LOG CAN AND CANNOT MEASURE

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ABSTRACT

Over a window of training, a neural network's parameter trajectory moves on a set far smaller than the parameter space. We call the dimension of that set the active dimension of the window. It counts independent components, and differs from three quantities it is often identified with: the number of directions the optimiser may move in, the rank of the map from those to the outputs, and the effective rank of the trajectory covariance. Measuring it directly means storing the trajectory; a scalar log costs nothing, so we ask how much of it survives the projection onto one. The estimator is standard, delay reconstruction followed by a maximum likelihood intrinsic dimension. It fails in one specific way: it returns a plausible number when the trajectory visits no invariant set. On six constructed systems, ending with a linear head on a frozen image classifier, it recovers the active dimension on withheld ranks and seeds to within about one component, up to a ceiling near eight directions. Two diagnostics computed from the series alone identify the regimes in which the estimate may not be read as a dimension; on grokking they place both standard settings outside the admissible regime. We therefore store the trajectory, whose effective rank collapses near generalisation in four regularised runs and then re-expands, the simplification that mechanistic accounts of grokking predict. Two runs trained without weight decay, which memorise and never generalise, decline as far over their whole budget, so the transition is marked not by the depth of the collapse but by its being sharp, locked to generalisation and reversed. Re-run at a window matched to that transition rather than to the record, the estimate on an ordinary log drops at the same step in the same four runs and in neither of the other two, by more than shape-preserving surrogates produce: two signatures of one event, the second already written by the run.

1 INTRODUCTION

A neural network is trained in a parameter space of dimension P, but the trajectory the optimiser follows does not fill it. Over a window it moves on a much smaller set, which grows and shrinks over a run. We call its dimension the *active dimension* of the window (equation (1)). It is a count of independent components: if the optimiser is carried by r independent quasiperiodic phases the orbit closes onto an r-torus and the active dimension is r, with no tunable scale. It is not the dimension of the subspace the optimiser is confined to, which the parameterisation fixes (Li et al., 2018). It is not the rank of the map from that subspace to the outputs. Nor is it the effective rank of the trajectory covariance, which anisotropy depresses below it. Section 3.1 separates the four.

Two questions are currently argued with whichever of those four quantities is least expensive to compute: whether an apparent simplification of training is a real reduction in degrees of freedom or an artefact of a shrinking scale, and whether a phase transition can be seen without instrumenting the run. Both are questions about the active dimension. Measuring it directly means storing the trajectory, which is expensive; a run already keeps several scalar logs for nothing, so we ask how much of the active dimension survives the projection onto one.

The estimator is standard: delay reconstruction (Takens, 1981; Sauer et al., 1991) followed by the pooled Levina–Bickel estimator (Levina & Bickel, 2004; MacKay & Ghahramani, 2005). It fails in one specific way, by returning a plausible number when the trajectory visits no invariant set. Power-law noise yields a finite reproducible dimension set by its spectral exponent (Osborne & Provenzale, 1989; Theiler, 1991), and a decaying transient, the path a converging run follows, yields a large one. Most of the work below therefore goes into establishing which case a given record is in, beginning with a validation protocol fixed before any result (section 4).

Contributions. Four quantities are frequently identified with one another: the dimension of the subspace the optimiser may move in, the rank of the map from it to the outputs, the number of independent components of the set it visits, and the effective rank of its covariance. We separate them, estimate only the third, and measure the fourth beside it everywhere so that the two cannot be conflated (section 3.1). We evaluate that estimate on six systems of increasing realism, ending with a constrained head on a network trained on image data: with no per-system calibration it recovers the active dimension on withheld ranks and seeds to within about one component, up to a ceiling we then locate (section 5). We establish what the absolute value depends on, and validate two diagnostics that separate the two failure regimes using the series alone (section 6). We then apply the whole procedure to grokking (section 7): the diagnostics place both published settings outside the admissible regime, the trajectory effective rank collapses at generalisation and re-expands, and at a window matched to that transition the log estimate declines with it in the runs that generalise and not in those that do not (section 7.3).

2 Related work

Reconstruction and intrinsic dimension. Delay reconstruction rests on Takens' theorem and its extensions (Takens, 1981; Sauer et al., 1991; Stark, 1999; Stark et al., 2003), is used on empirical series (Sugihara & May, 1990; Sugihara et al., 2012), and takes its lag and dimension from mutual information, false nearest neighbours or singular spectra (Fraser & Swinney, 1986; Kennel et al., 1992; Cao, 1997; Broomhead & King, 1986). We use pooled maximum likelihood (Levina & Bickel, 2004; MacKay & Ghahramani, 2005), not the alternatives (Grassberger & Procaccia, 1983; Facco et al., 2017; Camastra & Staiano, 2016); a finite record bounds them all (Eckmann & Ruelle, 1992). Oversampling (Theiler, 1986) and power-law noise (Osborne & Provenzale, 1989) produce spurious dimensions; surrogate tests (Theiler et al., 1992; Schreiber & Schmitz, 1996) detect nonlinear structure but not its amount, missing the transient regime entirely.

Dimension in optimisation. Random-subspace training (Li et al., 2018; Aghajanyan et al., 2021) reports the smallest k that still solves a task, a property of the task and not of a window; we borrow the parameterisation and measure something else inside it. Closest to the active dimension is the finding that descent occurs largely in a small subspace spanned by the top Hessian eigenvectors (Gur-Ari et al., 2018; Sagun et al., 2017). Ansuini et al. (2019) and Pope et al. (2021) estimate the intrinsic dimension of a data set or of a layer's representations. Those are properties of a point cloud; the active dimension is a property of a path. The participation ratio we adopt (Gao et al., 2017) is not the spectral-entropy quantity of the same name (Roy & Vetterli, 2007). We know of one prior measurement of it, an incidental one: nearly all of a grokking trajectory's variance lies in two principal components (Notsawo Jr. et al., 2023). All of these quantities need weights, gradients or activations; the nearest precedent for a scalar computed during training and acted on is the gradient noise scale (McCandlish et al., 2018). Mini-batch descent near a minimum is modelled as a stochastic process (Mandt et al., 2017), our third regime; we measure whether the edge of stability (Cohen et al., 2021) supplies the recurrence the first regime needs (Appendix Q).

Grokking. Delayed generalisation (Power et al., 2022) is studied in two settings and we use both. The first is regularised and, in our version, stochastic: the transformer configuration of Nanda et al. (2023), also used by Liu et al. (2023), on modular addition and on the S_5 task of Power et al. (2022). The second is the unregularised full-batch gradient descent of Gromov (2023), whose closed form supplies an analytic reference, on the modular polynomials and representability criterion of Doshi et al. (2024), which supply label-matched pairs; Appendix O states where each departs from its source. Explanations are framed in weight or activation space (Nanda et al., 2023; Varma et al., 2023; Liu et al., 2022; Barak et al., 2022) or in a lazy-to-rich transition (Kumar et al., 2024). Closest to us, Notsawo Jr. et al. (2023) predict grokking from early loss oscillations (Thilak et al., 2022) and Merrill et al. (2023) track competing subnetworks. We estimate a dynamical quantity with an independently measured ground truth.

3 Problem statement

3.1 Four quantities, and the one we estimate

A *window* $\mathcal{W}$ is a block of L consecutive optimiser steps. Everything below is defined on one window, and a run is described by a sequence of them, because what the optimiser is doing changes over training. Let $\boldsymbol{\theta}_t \in \mathbb{R}^P$ be the parameter vector at step t, confined by construction or by the dynamics to an affine subspace $\boldsymbol{\theta} = \boldsymbol{\theta}_0 + \boldsymbol{V}^\top \boldsymbol{c}$ with $\boldsymbol{V} \in \mathbb{R}^{k\times P}$ orthonormal, so that $\boldsymbol{c}_t \in \mathbb{R}^k$ holds the coordinates that can move. Three dimensions can be defined on $\mathcal{W}$. They are different numbers.

The *available* dimension is k: how many directions the parameterisation permits. The *functional* dimension is the rank of $\partial \boldsymbol{f}(\boldsymbol{c})/\partial \boldsymbol{c}$, with $\boldsymbol{f}$ the outputs on a fixed probe set; it bounds how many of those k directions can change the computed function, and we report it rather than estimate it. The *active* dimension is the subject of this paper: how many independent directions the optimiser actually explores inside the window. Let $\mathcal{M}_\mathcal{W} = \{\boldsymbol{c}_t : t \in \mathcal{W}\}$ be the set of points the trajectory visits there, and

$$d_{\mathrm{act}}(\mathcal{W}) \;=\; \dim_\varepsilon \mathcal{M}_\mathcal{W} \tag{1}$$

its box-counting dimension at resolution ε: cover $\mathcal{M}_\mathcal{W}$ with boxes of side ε, and d_{act} is the exponent d for which the number needed grows like ε^{-d}.

Two features of that definition are deliberate. First, ε *stays finite* where the textbook definition would let it go to zero. It has to: the estimator sees the set only through distances between nearby samples, so it can report the dimension only at the scale of its own neighbourhoods. Second, *the window must be long*: any orbit looks like a curve over a short enough stretch, so $d_{\mathrm{act}} = 1$ whatever the motion unless $\mathcal{W}$ outlasts the time the trajectory takes to return near itself.

The validation systems build the case $\boldsymbol{c}_t = F(\varphi_1(t), \ldots, \varphi_r(t))$, with r independent quasiperiodic phases: the orbit closes onto an r-torus and $d_{\mathrm{act}} = r$ at every ε below the weakest mode's amplitude. Above that scale the weakest modes are invisible and d_{act} declines, which is a fact about the set at that resolution rather than an error (section 6.4).

A fourth quantity is measured throughout and is not the active dimension. For a non-negative spectrum $\lambda_1, \ldots, \lambda_n$ the participation ratio

$$\mathrm{PR}(\lambda) \;=\; \frac{\left(\sum_i \lambda_i\right)^2}{\sum_i \lambda_i^2} \tag{2}$$

equals m exactly when m of the λ_i are equal and the rest vanish. On $\operatorname{spec}\operatorname{Cov}\{\boldsymbol{c}_t : t \in \mathcal{W}\}$ it gives the *effective rank* $\mathrm{PR}^{\mathrm{pos}}(\mathcal{W})$, the sense that term carries below, with $\mathrm{PR}^{\mathrm{det}}$ the same after the per-window linear detrend a drifting trajectory needs and $\mathrm{PR}^{\mathrm{upd}}$ the same on the increments. The effective rank is not that dimension: it equals r when r directions carry comparable variance and falls below r when they do not, while d_{act} stays at r (section 6.4). We use it to check that a construction excites the r directions it claims, and section 7.2 measures it on a real run. The four quantities come apart in practice: in section 5.4 an intervention that leaves the available and functional dimensions and the injected rank untouched still lowers the effective rank (table 15).

3.2 Observation model and estimator

We observe a scalar series $x_t = \phi(\boldsymbol{s}_t)$, with $\boldsymbol{s}_t$ the full optimiser state and ϕ fixed before the run, and reconstruct $\boldsymbol{y}_t = (x_t, \ldots, x_{t-(E-1)\tau}) \in \mathbb{R}^E$. For a generic pair of dynamics and observation on a compact invariant manifold of integer dimension d this map is an embedding once $E > 2d$ (Takens, 1981); the prevalence form replaces d by the box-counting dimension, which need not be an integer, and adds conditions on periodic orbits (Sauer et al., 1991). Genericity is not verifiable for a ϕ fixed in advance, so every result below is reported over a family of observers. We estimate the intrinsic dimension of $\{\boldsymbol{y}_t\}$ by maximum likelihood on nearest-neighbour distances (Levina & Bickel, 2004). Let $r_1^{(i)} \leq \cdots \leq r_m^{(i)}$ be the distances from point i to its m nearest neighbours, subject to a Theiler exclusion W_T on their separation in time (Theiler, 1986; Kantz & Schreiber, 2004). Pooling the

Table 1: The three regimes, the active dimension of equation (1) in each, and the value the estimator returns on the systems of section 5. The last two rows fail differently: on a transient the estimand exists and the protocol misses it, while under stochastic forcing there is no estimand to miss.

regime	what the orbit fills	d_{act}	estimate here
deterministic, recurrent	an r-torus	r	r
deterministic, transient	a curve, never revisited	1	≈ 29, at every r
stochastically driven	no low-dimensional invariant set	none	set by $E_{\max}$, not by r

likelihood over the N points before inverting, as MacKay & Ghahramani (2005) argue one should since the reciprocal is the unbiased quantity, gives

$$\hat{d}_{\mathrm{MG}}^{-1} \;=\; \frac{1}{N(m-1)} \sum_{i=1}^{N} \sum_{j=1}^{m-1} \log \frac{r_m^{(i)}}{r_j^{(i)}}, \tag{3}$$

while averaging the per-point estimates instead, as Levina & Bickel (2004) propose, gives the variant we write LB. We write *the estimate* for the value MG returns and *the estimator* for the procedure; table 9 reports both and Appendix A gives the algorithm as implemented. Three further statistics accompany every estimate, each sensitive to an effect that can produce a dimension signal with nothing geometric behind it: the *roughness* $\operatorname{std}(\Delta x)/\operatorname{std}(x)$, the *linear participation ratio* $\mathrm{PR}_{\mathrm{delay}}$ of equation (2) on $\operatorname{Cov}\{\boldsymbol{y}_t\}$, and the *spectral participation ratio* on the power spectrum of x.

3.3 Dynamical regimes

Two conditions must hold. They are not the same. The theorems require a deterministic flow on a compact invariant set, which licenses the delay map as an embedding. The neighbour statistic needs more: the record must visit that set densely enough for a point's nearest neighbours to be returns rather than its own temporal neighbours, which the Theiler exclusion enforces and section 3.4 tests. Recurrence enters here and only here: not as a requirement of the estimator, which would accept points drawn from independent runs, but as a requirement of a protocol whose entire sample is a single trajectory. Neither condition is needed for the optimiser to have an active dimension; both are needed to read it off one scalar record. An optimiser near a solution may be in any of the three regimes of table 1, and only the first meets them.

Mini-batch gradient noise produces the third regime, which the theory covers less well than it first appears. The embedding theorems for forced systems (Stark, 1999; Stark et al., 2003) cover it in part: the update may depend on the state arbitrarily, while their iterated-function-system form admits a driver selecting among finitely many maps, as sampling a finite dataset does. But they deliver an embedding *fibre by fibre*, one per realisation of the drive, and a single run crosses fibres, so no invariant measure exists there to sample. The second regime is less often noticed: a decaying transient traces a curve it never returns to, so the record offers the neighbour statistic no returns to work with; its effective rank remains near unity (table 15) only because that curve is nearly straight.

3.4 Admissibility diagnostics

Two statistics computed from the series alone separate the two failure regimes. The **identifiability ratio** $\rho_{\mathrm{ident}} = \hat{d}_{\mathrm{MG}}(2E_{\max})/\hat{d}_{\mathrm{MG}}(E_{\max})$ is near unity when a dimension is resolvable and grows when the estimate is a property of the embedding space rather than of the data, which is the stochastic regime. On a transient $\rho_{\mathrm{ident}} \approx 1$ as well, for the wrong reason, and the **trend-crossing count** distinguishes the two: the number of sign changes of the window's residual about its least-squares line, which is near zero there. It tests non-monotonicity of the series, not recurrence in the reconstructed space, so the two must be used together. A third statistic, the **degeneracy indicator**, identifies no regime. It is raised when too many neighbour distances or log-ratio sums reach their numerical floors, and guards against quantised or constant observers (Appendix A). Reference values are in section 6.4.

4 Validation protocol

Six requirements were fixed before any result. Each is named, so that table 5 can say which a system fails.

(1) Constructed truth. Every system is built with r independent quasiperiodic phases, so $d_{\mathrm{act}} = r$ by construction, and the recorded trajectory's effective rank confirms that all r are comparably excited.

(2) Frozen configuration. The estimator is chosen once, by a search over estimator settings only. Searching the system's settings at the same time — the drive period, say — would choose the data along with the method: the winning cell would report how favourable a system we found rather than how well the estimator works.

(3) Disjoint split. Ranks are withheld as well as seeds. Where the excitation geometry depends on r, every seed at a given rank shares that geometry, so withholding seeds alone would leave the rank itself in the training set.

(4) Zero learning rate. Re-run each system with the learning rate set to zero. The parameters then never move, so an observer of the optimiser must be flat: a constant log is exactly the right answer here. If instead the log keeps varying and returns the same estimate as before, the observer was never reading the optimiser. That happened in two of our systems: the drive acts on the targets, so the loss continues to move through the residual while the parameters are fixed.

(5) Companion statistics. The roughness, the linear participation ratio and the spectral participation ratio of section 3.2 are reported with every estimate, and where one of them does as well as the estimator we say so.

(6) Matched bandwidth across ranks. In a construction where each added direction also adds a higher frequency, the roughness alone orders the rank, so no embedding is being tested.

5 Evaluation on systems with a known active dimension

We evaluate the estimator on six systems of increasing realism: an oscillating diagonal matrix, online linear and logistic regression, a frozen nonlinear decoder, a perceptron in a k-subspace, and a linear head on a network trained on image data. In each of them the orbit closes onto an r-torus by construction, the measured effective rank confirms that all r directions are excited, and the estimator sees one scalar series. Table 5 collects the outcome.

5.1 An oscillating diagonal matrix

The first system has no optimiser and no feedback, so a negative result there would be decisive. In $\boldsymbol{W}(t) = \mathrm{diag}(b_i + \delta_i(t))$, r coordinates oscillate at rationally independent frequencies over non-zero offsets. The estimate tracks r across the whole range on both held-out seeds (table 5). It also separates geometry from smoothness. If several coordinates are driven from a single phase, the set they trace has one degree of freedom rather than several, and is no smoother for it. The estimate separates that case from the same number of independent oscillators by a factor of several, while the roughness of the series distinguishes them by a few per cent.

5.2 The ceiling

The estimator saturates above about eight directions. Varying the embedding dimension against the record length separates two quantities we had treated as one (Appendix R). The *level* returned at high rank belongs to the embedding: widening the delay window raises it, lengthening the record does not. The *rank up to which the estimate still tracks the truth* belongs to neither: it never passes about ten components, reaching that at the *shortest* record examined.

Both candidate explanations therefore fail. A finite-record bound (Eckmann & Ruelle, 1992) would allow far fewer components than we recover, and the embedding condition $E > 2d$ has the right direction but the wrong form: the ceiling climbs with $E_{\max}$ an order of magnitude more slowly than that rule requires, so its agreement with $E_{\max}/2$ holds only at the value we froze. One caveat is substantial: a purely linear statistic of the delay cloud follows the same curve, so what saturates may be the number of components the delay window resolves linearly rather than anything geometric.

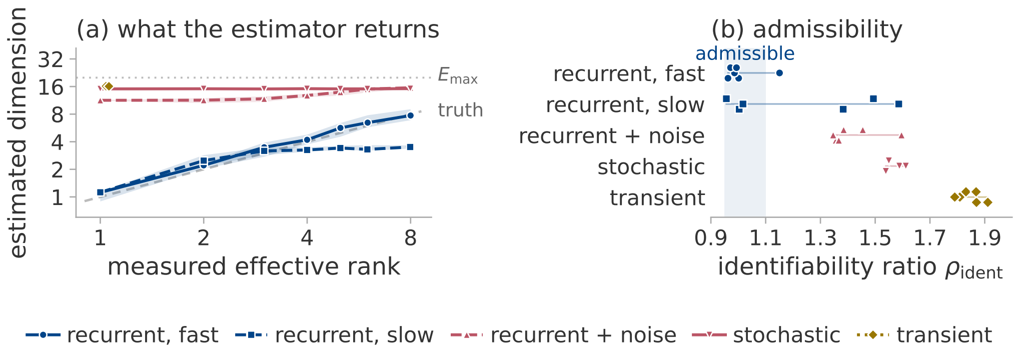

Figure 1: Recovery of the active dimension in each regime, and separation of the regimes without ground truth. Five of the nine excitation cases of section 5.4; colour is the regime of table 1. (a) The estimate against the measured effective rank, which the construction holds at r; the marker is a median over four seeds and ten observers, the band their interquartile range, the dashed diagonal the truth. The dotted line is $E_{\max}$, a reference and not a bound. (b) The identifiability ratio, on a smaller grid; the fast recurrent regime is the one it passes and the transient is furthest outside.

5.3 The zero-learning-rate test

Requirement 4 asks that a zero learning rate silence the observer; most published analyses of dimension on training logs do not run it. It invalidated two of our six systems: read naively they succeed, but they reproduce their reported results with the parameters frozen. It also rejects one observer, the instantaneous mini-batch loss of section 5.4, whose estimate goes on tracking the nominal rank when the parameters cannot move, though it is otherwise among the more accurate observers. Two further systems fail requirement 2 (table 5, Appendix D).

5.4 A constrained head on a network trained on image data

The sixth system uses real inputs: a tanh network trained on the scikit-learn handwritten digits and frozen, with a linear head confined to a k-dimensional affine subspace under softmax cross-entropy, so the curvature is data-dependent and all the dynamics occur in $\boldsymbol{c}$. The excitation is made r-dimensional by modulating the loss weights of r of twelve fixed data groups and orthogonalising the resulting gradient directions (Appendix F); the measured effective rank then matches r (table 15). Nine excitation cases are run, spanning the three regimes of table 1, and figure 1 shows five of them.

We record twelve observers (Appendix B). The estimator's parameters were frozen on four of them and on reserved seeds and ranks, so the results below are out of sample in rank and seed throughout, and in observer for most of what is scored. In the deterministic recurrent regime the estimate recovers the active dimension to within about one component, and every observer orders the withheld ranks correctly (table 5; Appendix D scores them one by one). The roughness of a log does not order the withheld ranks for any of the ten, so a smoothness explanation of that agreement is excluded throughout, the two parameter norms included. A configuration with twenty available directions gives the same picture at a lower ceiling. It is exploratory: it has no seed split, a Theiler exclusion smaller than its own embedding span, and $E = 2d$ exactly, one short of the condition in section 3.2 (Appendix N).

6 Conditions of validity

6.1 Dynamical regime

Figure 1 evaluates the estimator in each regime, at one nominal rank, observer and matched octave. Under stochastic excitation the estimate is flat in the rank and well above the true value, responding to $E_{\max}$ and the record rather than to anything in the data. Weak noise on an otherwise recurrent trajectory is less damaging: it destroys the absolute value and leaves the ordering.

A decaying transient returns near no state it has left, so the set it visits is a curve and $d_{\mathrm{act}} = 1$. The estimate exceeds that by more than an order of magnitude (table 1), and the Theiler exclusion is why. Remove it and the estimator returns, at every rank, seed and observer, the value maximum likelihood must return on a uniformly sampled curve, while the recurrent cases do not move (Appendix P). That value is not a level: it is a divergent quantity read at the exclusion cap, and its exceeding $E_{\max}$ is the expected behaviour of a maximum likelihood fit whose model is violated, not the failure of a bound. No clamping is applied (Appendix A).

Every system here is driven and training is not, so we test the one undriven candidate the literature offers: the edge of stability (Cohen et al., 2021). The perceptron of section 7 reaches it, oscillating while the loss rises on half of all steps. Oscillation is not recurrence, however: three of the eight runs at the edge oscillate at that rate and never return near a state they have left. The diagnostics do not tell the two groups apart, rejecting all eight (Appendix Q).

6.2 The delay lag

The delay window spans $(E_{\max} - 1)\tau$ samples, and must cover a real fraction of the oscillation period; otherwise the delay coordinates are collinear and the torus never unfolds. This is a property of the delay window rather than of the lag alone, which is standard (Casdagli et al., 1991). On a fixed torus, varying τ alone moves the estimate over more than an order of magnitude, passing through the truth only where the delay window covers a few tenths of a period (table 11, figure 4).

The right lag follows from the oscillation period, which a controlled experiment knows and a training log does not. We do not solve this. The lag used in section 7 is the frozen one, transplanted from a calibration system whose drive period is far shorter than the autocorrelation time of those logs, which by itself forbids reading an absolute value there. The identifiability ratio inherits the problem: doubling $E_{\max}$ doubles the delay span, so a large value means either that the estimate belongs to the embedding space or that the lag is mismatched, and on the grokking logs both readings are live.

6.3 Nuisance factors

At a fixed active dimension of four we vary seven factors that leave it unchanged, over five seeds and twelve observers, and compare each against a real four-component change (table 12). Two of them matter. Smoothness is not one of them: shifting the drive band by an octave changes the roughness sharply and moves the estimate by well under a fifth of what a real four-component change moves it, on every observer. The second factor is the observer's gain. A constant rescaling leaves the estimate unchanged, the scorer standardising it away, but a *ramped* gain moves it by about a component, so the estimate on a run whose parameter norm grows will drift by that much even when nothing dimensional happens. That is the nuisance that matters most in section 7.

Roughness tracks the estimate closely enough here, and again in section 7.3, to raise a prior question: are the two statistics separable at all? Appendix S answers it in both directions. Collapsing a log's four independent phases into one and then restoring them, with roughness held fixed by construction, leaves the estimate following the count $4 \to 1 \to 4$. Passing a single four-phase system through five monotone observers changes roughness by more than four fifths without changing any intrinsic dimension, and the estimate stays at four. The two statistics are therefore separable in both directions, on the parameter norm as on the rest.

6.4 The estimand is a dimension, not an effective rank

Every construction in section 5 equalises the drive amplitudes, which makes the active dimension and the effective rank of section 3.1 numerically equal, and an experiment in which two candidate estimands agree cannot say which the estimator reports. Driving mode l at amplitude q^l separates the two: the orbit still closes onto an r-torus, so $d_{\mathrm{act}} = r$, while the effective rank decays with q.

The estimate follows the dimension (table 14, figure 5): as the drive is made anisotropic the effective rank decays while the estimate stays with the number of modes, and across the grid it is closer to that number than to the effective rank. It registers only the modes it can resolve: once the weakest lies far enough below the strongest, the estimate decays with the rank, because the torus it can see has fewer dimensions than the one constructed.

The diagnostics. With no ground truth, the identifiability ratio separates the accurate cases from the inaccurate ones without overlap, within a regime as well as between them. It is not sufficient alone — on a transient $\rho_{\text{ident}} \approx 1$ as well, for the wrong reason — and must be paired with the trend-crossing count, which is near zero there. Both are calibrated on these five cases only, which is why section 7 treats the admissible band as a description and not a decision rule.

7 Application to grokking

We work in both settings of section 2, which differ in every relevant respect (Appendix O lists every run and every departure from the sources). The transformer runs are regularised, trained with AdamW under cross-entropy on mini-batches, though the source is full-batch, which puts them in the stochastic regime of table 1. The perceptron runs are unregularised full-batch descent on mean squared error; we take the tasks and representability criterion of Doshi et al. (2024) but not their optimiser (Appendix M). In both settings the diagnostics reject the level the logs report, so we store the trajectory and measure it directly, then return to the logs with the one question a rejected level still permits.

7.1 Neither setting is admissible

The frozen configuration of section 5.4, with only the window and stride shortened to fit these records, gives table 16 and figure 6. The two settings fail in different ways. Neither failure appeals to any label. The regularised mini-batch runs occupy a band of the identifiability ratio the deterministic regimes never reach, and their estimates scatter widely across seeds of one configuration, set by $E_{\max}$ and the record rather than by any rank, exactly as figure 1 predicts for stochastic excitation. The full-batch runs carry a transient's signature instead: $\rho_{\text{ident}} \approx 1$ with almost no trend crossings. In neither setting may the value be read as a dimension.

A change inside a run might still be read, since the offset a fixed configuration imposes is constant within a run, and the floor is the drift a ramped gain alone produces (section 6.3). That is not possible at this window, which spans tens of thousands of optimiser steps against a transition hundreds of steps wide; the limitation is resolution and not signal, so section 7.3 shortens the window (Appendix L).

Label-matched pairs. The perceptron setting supplies the comparison the first cannot: both members of a pair fit the training set completely, under identical optimisation, and differ by one monomial. The estimate is higher for the non-generalising member in every pair, with no overlap between the groups. It is still not a dimension: all eight runs are transient, the separation is already visible in the raw loss curves, the direct measurement does not reproduce it, and a decline is not specific to generalisation — one occurs in a long run that never leaves chance accuracy (Appendices G, J and K).

7.2 The trajectory simplifies at generalisation

A stored trajectory supports a measurement a log cannot: the effective rank of section 3.1 is defined whether or not the orbit recurs. We sketch it in parameter and in function space at every logged step of a transformer trained to grok (Appendix I). Near generalisation it collapses in both spaces, in parameter space to near unity, each within a few hundred steps of the transition, and both then re-expand (figure 2, table 17). Two observations support that reading. The collapse occurs at generalisation in seeds whose generalisation steps differ several-fold, and in a second task, so it does not track training progress. The displacement reaches its floor thousands of steps later and never recovers, whereas the rank does, so the trajectory has not simply stopped moving. The effective rank measures variation inside the window rather than the extent of the set the orbit visits, so the collapse is in that variation, not in d_{act}.

The runs without weight decay. The same two configurations trained without weight decay memorise and stay at chance accuracy. In the transition window they decline much less than the generalising runs, but over their whole budget one of them declines just as far (table 17): depth does not separate the groups, timing and shape do. Both peak early and then settle, and their parameter norm never returns to its earlier level, whereas every generalising run gives back much of its growth.

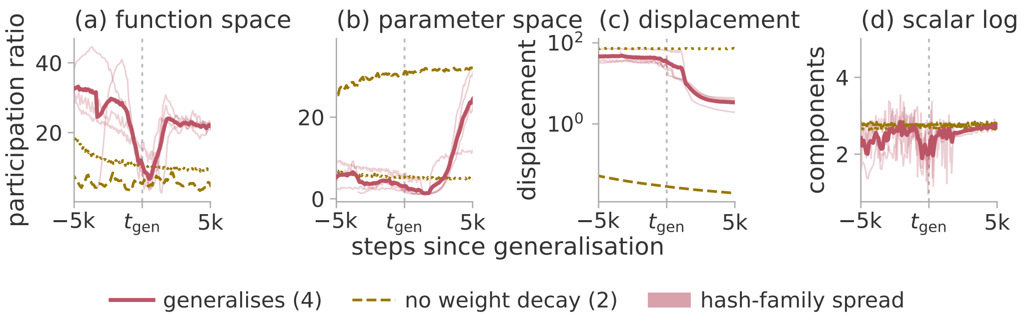

Figure 2: Simplification of the trajectory at generalisation. Three regularised seeds of modular addition and one S_5 composition, against the same two configurations without weight decay, each aligned on the generalisation step of the run it matches; colour is the regime of table 1, the thick line the mean of the four that generalise. The statistic in (a)–(c) is $\mathrm{PR}^{\mathrm{det}}$, an effective rank: none of these runs recurs, so none supports a dimension. Windows are 600 optimiser steps, 300 for S_5, and the band on (a) and (b) is the two-hash disagreement (Appendix I). Panel (d) is a different measurement at the same midpoints, the delay estimate of section 7.3 on the parameter-norm log.

A low rank is reached both by a run that has come to rest and by one about to generalise; the growing norm is the ramped gain of section 6.3, the mechanism behind the decline of section 7.1.

7.3 The log estimate falls where the trajectory does

Setting the window in units of the transition rather than of the record makes the test of section 7.1 possible. We re-run the estimator on the same logs, at the window and midpoints the direct measurement uses, so both statistics are sampled at the same instants. The frozen configuration will not fit a window that short, and requirement 2 forbids choosing a replacement on the outcome, so we report every cell of a grid and take as headline the one a rule blind to the outcome selects (Appendix L).

On the parameter norm the estimate halves at the generalisation step in all four generalising runs, its floor on the same side of the transition as that of the direct measurement (figure 2d). That drop is among the largest each generalising run ever shows and is unremarkable in the other two, and all but one usable grid cell separates the groups (table 20). It is not an artefact of the observer's shape, the nuisance factor of section 6.3: against surrogates that keep the log's shape and its fluctuation spectrum, every generalising run's p lies below both of theirs, at every smoothing length and seeding tried (Appendix L).

These are two signatures of one event: one obtained by storing the trajectory, one already written by the run. Three qualifications keep the second modest. Its level cannot be read as a dimension, the identifiability ratio being uncomputable here, so only the change is read. The observer is the parameter norm, for which section 6.3 cannot exclude a smoothness explanation, so the surrogate test carries the claim. A simpler statistic registers the change more strongly: the roughness of the same windows collapses by an order of magnitude while the estimate merely halves, so most of what this log says about the transition is in that collapse, which needs no embedding.

8 Conclusion

In a deterministic recurrent system with a matched delay lag, one scalar log of a suitable observer counts the independent components of the set the optimiser visits, to within about one component and up to about eight. Outside that regime it does not: the set a transient visits is a curve, and stochastic forcing leaves no invariant set at all. The estimator is therefore usable only alongside the two diagnostics, which on grokking place both standard settings outside the admissible regime. Measured directly, the trajectory effective rank collapses at generalisation and re-expands, while the two runs without weight decay move much less, and at a matched window the log estimate declines with it.

Limitations. The measurement needs a recurrent regime and a lag matched to a period no log reveals; above about eight components it is only ordinal, the configuration being a choice as much as a measurement (Appendix C). The collapse rests on four runs against two others differing in the variable that also drives the parameter norm; it is a signature, not an early warning, and does not reproduce in the second setting. The matched-window drop is close to what a ramped gain alone produces, one of those two has little range to move through, and the edge-of-stability evidence is seven runs on one task (Appendices J, L and Q).

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A THE ESTIMATOR

Algorithm 1 is the estimator as it was run. It computes equation (3) with three departures that affect what is reported. It also states the constants the main text leaves unnamed.

Algorithm 1 Pooled Levina–Bickel intrinsic dimension of a delay reconstruction

Require: series $x_{1:n}$; lag τ; embedding dimension E; neighbours m; Theiler exclusion W_T; dither σ

$z_t \leftarrow (x_t - \operatorname{mean} x)/\operatorname{std} x + \sigma\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0,1)$ ▷ breaks exact ties
$\boldsymbol{y}_t \leftarrow (z_t, z_{t-\tau}, \ldots, z_{t-(E-1)\tau})$ for $t = (E-1)\tau + 1, \ldots, n$
$S \leftarrow 0$; $N \leftarrow 0$; $n_{\text{dist}} \leftarrow 0$; $n_{\text{ratio}} \leftarrow 0$
for each $\boldsymbol{y}_i$ **do**
    $r_1^{(i)} \le \cdots \le r_m^{(i)} \leftarrow$ distances to the m nearest $\boldsymbol{y}_j$ with $|i-j| > W_T$
    $r_j^{(i)} \leftarrow \max(r_j^{(i)}, \varepsilon_{\text{dist}})$ for each j; count those clipped into n_{dist}
    $S_i \leftarrow \max\big(\sum_{j=1}^{m-1} \log(r_m^{(i)}/r_j^{(i)}),\ \varepsilon_{\text{ratio}}\big)$; count clips into n_{ratio}
    $S \leftarrow S + S_i; \quad N \leftarrow N + 1$
end for
mark the window degenerate if n_{dist} or n_{ratio} exceeds $1\,\%$ of its denominator
return $\big(N(m-1)-1\big)/S$

The three departures are as follows. *The returned value is* $(N(m-1)-1)/S$ *rather than* $N(m-1)/S$, the exactly unbiased pooled maximum likelihood estimate for a $\Gamma(N(m-1), 1/d)$ sample, which differs from equation (3) by less than 10^{-5} at the sizes used here. *A window marked degenerate is returned rather than discarded*: the constructed-system pipelines drop marked windows when summarising a series over its sliding windows, the two grokking pipelines take the median over all of them, and no published cell of table 16 contains one. *The returned value is never clamped to* E, so a divergent estimate is reported as divergent; clamping would convert the transient regime of table 1 into a plausible number.

The floors are $\varepsilon_{\text{dist}} = 10^{-8}$ on a neighbour distance and $\varepsilon_{\text{ratio}} = 10^{-5}$ on the per-point log-ratio sum, with $\sigma = 10^{-9}$ on the standardised series. The Theiler exclusion is $W_T = \max\big((E-1)\tau,\ t_{\text{acf}}\big)$, where t_{acf} is the first lag at which the autocorrelation falls below $1/e$, capped at 150 samples since the neighbour query cost grows linearly with it. Appendix N measures the effect of that cap.

B THE OBSERVERS

Twelve scalar observers are recorded on the system of section 5.4, in five families; table 2 defines each. Two are excluded from every aggregate, for the reasons appendix D gives. "Parameter norms (two)" in table 7 is the pair $\|\boldsymbol{\theta}\|$ and $\|\boldsymbol{c}\|$, reported as one row because their errors agree to two decimals; they are distinct quantities and both are given here.

Table 2: The twelve observers. $\boldsymbol{c}$ is the coordinate in the k-dimensional subspace and $\boldsymbol{\theta}_0$ the frozen head at initialisation; $\boldsymbol{L}$ is the matrix of probe logits, where the probe is a fixed held-out set drawn once before training; $\boldsymbol{g} = \partial\ell/\partial\boldsymbol{c}$. The projection directions $\boldsymbol{u}$, $\boldsymbol{v}$ and $\boldsymbol{a}$, and the rotation $\boldsymbol{R}$, are drawn once from a fixed seed and never redrawn.

name	family	definition
instantaneous loss	loss	the loss on the current minibatch
full-batch loss	loss	the loss on the whole training set
probe loss	loss	cross-entropy on the probe set
parameter norm	norm	$(\lVert\boldsymbol{\theta}_0\rVert^2 + \lVert\boldsymbol{c}\rVert^2)^{1/2}$
subspace norm	norm	$\lVert\boldsymbol{c}\rVert_2$
function-space norm	norm	$\lVert\boldsymbol{L}\rVert_F$
gradient norm	gradient	$\lVert\boldsymbol{g}\rVert_2$
gradient projection	gradient	$\langle\boldsymbol{u}, \boldsymbol{g}\rangle$
fixed parameter projection	projection	$\langle\boldsymbol{v}, \boldsymbol{R}\boldsymbol{c}\rangle$
function-space projection	function	$\langle\boldsymbol{a}, \operatorname{vec}\boldsymbol{L}\rangle$
margin	function	mean probe logit gap, true class less best other
probe accuracy	function	accuracy on the probe set

C FROZEN CONFIGURATIONS

Two estimator configurations are used in this paper, one for each range of the active dimension. Both were selected once, on data withheld from every later experiment, and then frozen. Table 3 states them and the grids they came from.

Table 3: The two frozen configurations and their selection. Only the swept parameters were varied; the remainder were fixed before any estimate was computed. Three settings are common to both and are not repeated: $m = 20$ neighbours, a Theiler exclusion of at least the embedding span (appendix A), and a dither of 10^{-9}; the first two were swept in the eight-direction grid, where the two Theiler rules return bit-identical values and the selection between them is arbitrary.

	eight-direction	twenty-direction
$E_{\max}$	20	40
τ	4	16
window / stride	8 000 / 2 000	8 000 / 4 000
swept	$E_{\max}$, τ, m, Theiler, window	$E_{\max}$, τ
grid size	48	4
selection seeds	90, 91, 92	0, 1, 2
selection ranks	2, 4, 6	2, 6, 10, 14, 18
withheld seeds	0–3	none
withheld ranks	1, 3, 5, 8	the other fifteen
selection error, best	0.17	2.47
selection error, worst	2.03	3.35

The selection objective in both cases is the error of the median estimate against the median measured effective rank, over the four observers reserved for selection, one from each observer family. The eight-direction objective adds a penalty on the rank correlation and on the spread across seeds; at the twenty-direction configuration both penalties are identically zero, so the objective there reduces to the error alone.

Five properties of these choices need to be stated.

Both lie at a grid boundary. The eight-direction configuration is at the top of all five swept axes and the twenty-direction one at the top of both. That is less serious than it appears for the first: four of its five axes have only two levels, so every configuration in that grid is at a boundary in four of them, and τ, the only axis with an interior value, has its winner at the top end. Even so, neither configuration is interior, and a wider grid could move the level.

The selection error spans a factor of twelve on identical data. Across the eight-direction grid the error runs from 0.17 to 2.03 on the same logs, so the level is a choice as much as a measurement. Only the withheld numbers of section 5.4 may be read as recovery.

The Theiler axis is inert on this data. The autocorrelation rule returns the larger of the embedding span $(E_{\max} - 1)\tau$ and the measured autocorrelation time. On these logs the autocorrelation time is one to three samples against an embedding span of nine or more, so the two Theiler settings return bit-identical values wherever they are compared and the 48-point grid contains 24 distinct measurements.

Neither configuration always uses the exclusion its parameters name. The implementation caps the Theiler exclusion at 150 samples, which binds unconditionally at the twenty-direction configuration and on five of the seven training logs of section 7. Appendix N states the departure and measures it.

The stride is not always the frozen one. Three pipelines override the window geometry and no estimator field; table 4 states what each of them uses. The constructed-system setting produced the 0.90 of section 5.4, so "at the frozen configuration" is true there of the estimator and not of the stride. Every other field — $E_{\max}$, τ, m, the Theiler rule, the dither — is used as frozen throughout. The division of the record into windows changes between experiments; the scoring of a window does not.

Table 4: Which window and stride each pipeline uses, and where they depart from the frozen configuration of table 3. Lengths are in samples of the log. The longer stride on the constructed systems keeps the number of windows near ten whatever window length was frozen, and the second value applies where the absent burn-in makes the record longer.

pipeline	window	stride
frozen configuration	8 000	2 000
constructed systems (section 5)	8 000	3 000, or 3 666
training logs (section 7)	a third of the record	1 000

D Per-observer results and alternative statistics

Table 5 gives the evaluation of section 5 in full, table 7 scores each observer on its own, and table 9 sets the estimator against every statistic we could compute on the same data. Two rows of table 5 carry a caveat and are taken first.

Table 5: How accurately the estimator recovers the active dimension on each system, and which requirements of section 4 that system fails. Systems are ordered by increasing realism. The unit of analysis is one (rank, seed, observer) run, each side of the error being the median over that run's seven sliding windows; ρ is over ranks after a median over seeds. The two oscillating-matrix figures are its two held-out seeds. The function-subspace row fails requirement 1 on its soft half only: its three highest ranks reach an effective rank of about $0.86\,r$, short of the $0.9\,r$ the check asks, while every hard rank is exactly r.

system	truth	range	MAE	ρ	protocol
oscillating matrix	constructed	1–10	0.30 / 0.69	1.00 / 0.99	fails 2, 5
online linear regression	constructed	1–20	0.79	0.99	fails 2
logistic regression	constructed	1–20	0.81	1.00	fails 2
frozen nonlinear decoder	constructed	1–20	1.58	0.94	fails 4
perceptron in a k-subspace	constructed	1–20	0.92	0.99	fails 4
image data, parameter subspace	measured	1–8	0.90	1.00	**complete**
image data, parameter subspace	measured	1–20	1.75	1.00	fails 3
image data, function subspace	measured	1–8	0.28	1.00	fails 1, 2, 3, 4

Two rows of table 5 predate the estimator of appendix A. The oscillating matrix was scored with an earlier implementation that applies no Theiler exclusion and clamps the returned value to E, at $E = 25$ and $m = 30$ rather than at the frozen configuration; its grid swept the drive period alongside the estimator parameters, which is the breach of requirement 2 that also disqualifies the two regression systems. It reports no companion statistics either, which is requirement 5. The function-subspace row shares the clamping and the mixed grid, computes one window per run rather than a median over sliding windows, and was never re-run at zero learning rate, so requirement 4 is untested there rather than passed.

Both rows are kept because they locate the ceiling, and because the oscillating matrix supplies the synchronisation test of section 5.1, which no other system provides. Neither should be read as a recovery result.

The two implementations differ in their defaults and not in their arithmetic: at matched settings they agree to 0.0 on all thirty cells of a validation set and on twelve further series, both calling the same neighbour kernel. Of the two defaults that differ, the clamp never fired: the largest estimate anywhere in these files is 22.34 against a threshold of 62, so it is a difference that never took effect rather than a correction that was applied. The missing exclusion is not free, but its cost is small and not consistently signed. Switching it on moves the held-out error of table 5 by about three hundredths, upward on one seed and downward on the other; on the first seed it also costs the perfect rank correlation.

All numbers in this appendix are computed on the withheld ranks $r \in \{1, 3, 5, 8\}$ and the withheld seeds of section 5.4, in the deterministic recurrent regime, and are scored against the measured effective rank rather than against r. Where an aggregation over seeds is needed, the median over the four seeds at each rank is taken before scoring; the choice moves the numbers, so it is stated. Three figures in this paper describe the same eight-direction measurement under three aggregations, and table 6 separates them; they should not be read as three results.

Table 6: Why three different errors are quoted for one measurement. Each row is the same eight-direction recovery of section 5.4 — withheld ranks, withheld seeds, deterministic recurrent regime — summarised in a different order, and the order is stated wherever one of these figures appears.

aggregation	reported in	error
mean over (rank, seed, observer) runs	section 5.4, table 5	0.90
median over observers of each observer's error	table 7	0.48
median over seeds and observers at each rank, then scored	table 9	0.382

Table 7: Which scalar observer to prefer, and whether its accuracy could have come from smoothness alone. One row per observer, except that the two parameter norms share one; MAE and slope are over the four withheld ranks, median over the four withheld seeds at each, scored against the measured effective rank. The rank correlation is 1.00 for every observer, so it has no column. The last column records whether the roughness statistic orders the four ranks perfectly, which with $n = 4$ can only be a yes or a no.

observer	family	MAE	slope	roughness orders rank
probe loss	loss	0.60	0.82	no
function-space norm	norm	0.39	0.96	no
margin	function	0.46	0.83	no
parameter norms (two)	norm	0.41	1.04	no
fixed parameter projection	projection	0.34	0.83	no
gradient projection	gradient	0.70	0.72	no
function-space projection	function	0.51	0.78	no
gradient norm	gradient	1.81	1.58	no
full-batch loss	loss	1.87	1.51	no
instantaneous loss	loss	0.61	0.73	no
probe accuracy	function	degenerate in every window		

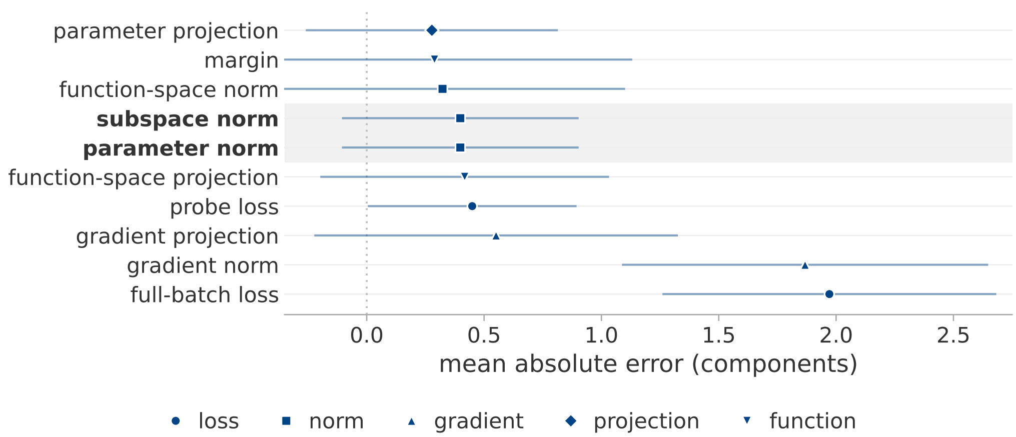

Figure 3: Which scalar log to keep. Ten of the twelve observers of table 2 on the fast recurrent case of section 5.4. The point is the mean absolute error against the measured effective rank over seven ranks, median over four seeds; the bar is the across-seed spread of the estimate and not of the error, so where it crosses zero the error is inside the seed-to-seed noise. Marker shape is the observer family and the shaded rows are the two parameter norms. Eight of the ten overlap, so this experiment does not choose among them; the gradient norm and the full-batch loss are far worse and should not be used. The ordering differs slightly from table 7.

Two observers are excluded from every aggregate, for different reasons. Probe accuracy is quantised, which collapses the delay vectors onto one another and raises the degeneracy indicator in $100\,\%$ of its windows; it is the sole source of the degenerate fraction reported anywhere in this study, every other observer being at zero. The instantaneous mini-batch loss is excluded because it fails requirement 4: at zero learning rate, where the parameters do not move at all, its estimate still climbs from 0.91 to 6.93 across the seven simulated ranks, a range of six components produced entirely by the drive. The other eleven observers are exactly constant at zero learning rate, so no estimate is defined for them.

The roughness column settles the smoothness question for this experiment. No scored row has a roughness statistic that orders the four withheld ranks, the best reaching a rank correlation of 0.8 and the two parameter norms 0.4, so smoothness alone accounts for no row's agreement with the truth. The published construction, whose drive geometry did not vary with the seed, ordered them perfectly on the two parameter norms; correcting the drive removed the coincidence along with the caveat it forced.

Table 8: How far recovery extends when twenty directions are available rather than ten. Image data, median over three seeds and eight observers; the truth is the measured effective rank, and the last row is the spread of MG across seeds at fixed rank.

true r	1	2	4	6	8	10	12	14	17	20
MG	1.10	2.05	4.51	6.51	7.45	9.92	10.23	11.40	13.90	15.44
$\mathrm{PR}_{\mathrm{delay}}$	2.00	2.26	3.79	5.42	7.90	9.45	9.43	11.25	11.43	13.42
spectral PR	1.09	1.27	2.13	2.89	4.10	4.60	5.51	6.22	6.81	7.48
spread across seeds	0.87	0.51	1.35	1.81	2.64	3.59	2.81	2.07	3.41	4.24

Table 9: Whether any simpler statistic recovers the active dimension as well as the estimator does. All are computed on the image-data system of section 5.4. The two halves are not aggregated alike: at eight directions each statistic is scored on the four withheld ranks, at twenty on all twenty, the five selection ranks included, so only the comparison down a column is out of sample. Statistics marked † need a neighbour search; the rest are one-line computations on the series or its spectrum. Correlations are given to two decimals at eight directions and to three at twenty, so that 0.997 is not printed as 1.00.

	eight directions		twenty directions	
statistic	MAE	ρ	MAE	ρ
MG †	0.382	1.00	1.75	0.997
LB †	0.40	1.00	1.52	0.997
TwoNN †	2.13	0.80	2.79	0.880
$\mathrm{PR}_{\mathrm{delay}}$	0.91	1.00	2.48	0.985
spectral PR, 256 bands	1.25	1.00	5.83	0.998
spectral PR, 1024 bands	1.08	1.00	4.78	0.979
spectral PR, native resolution	0.69	1.00	2.95	0.976
roughness	3.65	0.20	9.89	−0.379

Three qualifications attach to table 9, and none of them favours MG.

First, MG is not the most accurate statistic at either range: the uncorrected per-point average LB is ahead of it at both. The correction of section 3.2 removes a positive bias, and on a saturating estimate that bias was partly cancelling the saturation, so removing it costs accuracy exactly where the estimate is already saturating.

Second, MG is not the best-ordering statistic at twenty directions. The spectral participation ratio at 256 and 1024 bands orders the ranks more faithfully than MG does, at three times the error. If only the ordering is wanted, a statistic requiring no embedding and no neighbour search will supply it.

Third, the advantage of MG over the linear delay participation ratio is a factor of about two at eight directions, not the factor of six that the selection split of appendix C reports. That larger figure is computed on the ranks and seeds the configuration was chosen on and is a property of the selection, not a recovery result.

Dependence on the observer. We run the same torus through two scalar observers that differ only in how evenly they weight the modes. Neither statistic is preferable in general (table 10). Under an observer that weights all modes equally the spectral participation ratio is the more accurate; under a generic observer with random weights — the form a random projection of a network's parameters actually takes — MG is. Both orderings are perfect in three of those four cells, so the statistics are separated by absolute recovery and not by rank agreement.

The size of each advantage also depends on the delay lag, which a controlled experiment can choose and a training log cannot. Held at a lag common to both observers rather than at the one best for each, the direction of each advantage survives and its size does not.

Table 10: Whether MG or the spectral participation ratio recovers the active dimension more accurately, and how far the answer depends on the observer and on the delay lag. MG against the spectral participation ratio at 256 bands, on one torus; each figure is the ratio of their errors, in favour of the statistic named in the second column. The first column of figures uses the lag best for that observer, which a training log cannot choose; the last two hold both observers at a common lag.

observer	more accurate	at its best lag	$\tau = 4$	$\tau = 1$
equal mode weights	spectral PR	3.4	6.6	3.4
random mode weights	MG	4.7	4.7	1.9

E CONDITIONS OF VALIDITY IN DETAIL

This appendix gives the three experiments section 6 summarises: the delay lag, the nuisance factors and the anisotropy grid. It adds one the main text does not report, a test that a change in the active dimension is detected when one occurs.

The delay lag. The delay window spans $(E_{\max} - 1)\tau$ samples, and table 11 varies τ on a fixed torus of period 400 at $E_{\max} = 20$, leaving the system unchanged. Below a tenth of a period every rank returns the same value near 2.5 and the ordering is lost; between three tenths and a half of a period the estimate follows the rank; above one period it diverges. The failure is a collapse of the ordering at one end and a divergence at the other, not a uniform bias.

Table 11: How far the estimate depends on the delay lag. A fixed torus of period 400 at $E_{\max} = 20$; the span is the delay window in units of the oscillation period. The correct rank is the column heading.

τ	span	$r = 1$	2	3	4	6	8
1	0.05	1.33	2.23	2.50	2.61	2.41	2.44
4	0.19	1.33	2.06	3.23	3.93	3.75	3.77
8	0.38	1.34	2.08	3.80	6.48	5.93	6.12
16	0.76	1.35	2.25	5.13	11.80	10.34	13.29
32	1.52	1.37	2.28	6.83	21.61	22.63	41.03

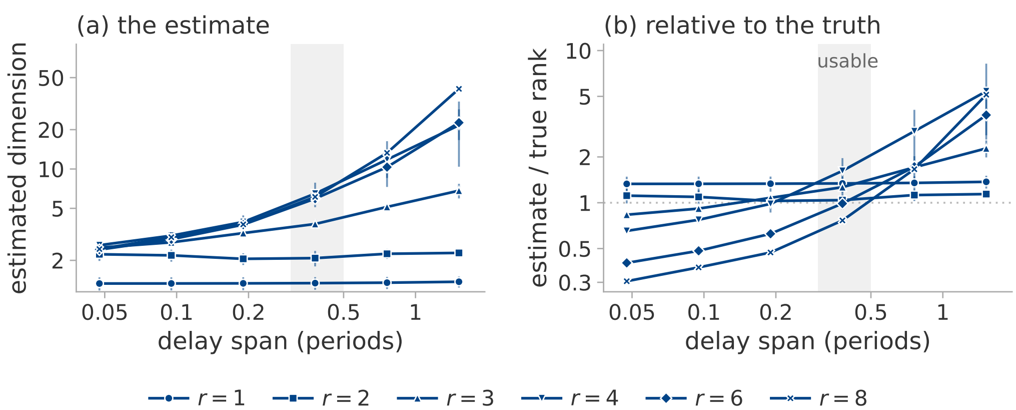

Figure 4: The delay lag, varied on one system: a fixed torus of period 400 samples at $E_{\max} = 20$, two seeds, six lags spanning 0.05 to 1.52 periods. Marker shape is the true rank, the marker the median of the two seeds and the bar their range. (a) The estimate. (b) The same divided by the true rank, so one reference line replaces six; the shaded column is three tenths to a half of a period. Table 11 gives the same numbers. A seventh setting, with τ taken from the autocorrelation, is omitted because its span differs between the seeds.

Nuisance factors. Table 12 varies seven factors that leave the active dimension at four. Rotating the coordinates is scored on the fixed parameter projection alone, since the other eleven observers are invariant to it by construction, so that row measures a property of the observer rather than of the estimator.

Table 12: Shift in the estimate produced by factors that leave the active dimension unchanged, median over five seeds and twelve observers. The first row is the seed-to-seed variability of an unchanged configuration and sets the scale for the others. The last column is a percentage of a real four-component change, which the change-detection experiment below observes as a fall of 3.96. A constant rescaling of the observer moves the estimate by 0.000 and is omitted.

factor	$\lvert\Delta\mathrm{MG}\rvert$	as % of a real four-component change
none (baseline)	0.39	10
learning rate halved	0.37	9
drive band up one octave	0.50	13
drive band down one octave	0.78	20
coordinates rotated	0.28	7
observer gain ramped 10×	1.53	38
state amplitude ramped 4×	1.57	39

Detection of a change. Recovery of a static level does not establish that a change is detected. We run three consecutive segments $r_{\mathrm{high}} \to r_{\mathrm{low}} \to r_{\mathrm{high}}$ with nothing else altered, re-measuring the ground truth on each. In the deterministic recurrent regime a true fall of 4.00 components is observed as 3.96, is detected in both directions on over nine tenths of the runs, and recovers to within 0.01 of the level before it. Under stochastic excitation the same true change is observed as 0.09.

The rule fires when the estimate falls below the median of the pre-switch segment by three times the window-to-window scatter of that segment, for two consecutive windows. The threshold uses the pre-switch segment alone, never the midpoint of the two observed levels: a midpoint would give the detector the answer, and would resolve the case of two coinciding levels arbitrarily.

Table 13 gives the three schedules on the parameter norm. The level error is 0.45 at the high level and 0.22 at the low one: the estimate is more accurate the fewer directions are active.

Table 13: Whether a change of the active dimension is recovered at both levels and after it reverses. Three schedules $r_{\mathrm{high}} \to r_{\mathrm{low}} \to r_{\mathrm{high}}$, on the parameter norm. At $8 \to 3 \to 8$ the high level is under-reported by a component, which is the saturation of table 8 seen from another direction.

schedule	high	low	high again
$4 \to 1 \to 4$	4.06	0.84	4.06
$6 \to 2 \to 6$	7.00	2.17	6.95
$8 \to 3 \to 8$	8.32	3.34	8.28

The response time cannot be quoted from this experiment. With a window of 8 000 samples and a stride of 2 000 the two observed lags, 999 and 1 999 steps, are the two floors of the measurement grid rather than measured values. The experiment does establish that the estimate crosses the threshold on a window still containing 87.5 % pre-switch data.

Anisotropy. Table 14 gives the measurement of section 6.4 in full: the system of section 5.1 with mode l driven at amplitude q^l, three seeds, observed through the squared Frobenius norm and estimated with the frozen eight-direction configuration. The manifold dimension is r in every row.

Table 14: Whether the estimate follows the number of modes or the effective rank when the two disagree. An r-torus with geometrically decaying amplitudes; r is the manifold dimension and is constant along each block, q the amplitude decay factor; PR is the participation ratio of the covariance of the r oscillating coordinates. Median over three seeds.

r	q	PR	MG	$\mathrm{PR}_{\mathrm{delay}}$
2	1.0	2.00	2.11	3.65
2	0.7	1.79	2.17	3.40
2	0.5	1.47	2.26	2.93
4	1.0	4.00	4.07	6.11
4	0.7	2.60	4.60	4.88
4	0.5	1.65	4.99	3.32
6	1.0	6.00	7.31	7.93
6	0.7	2.84	6.97	4.38
6	0.5	1.67	4.48	3.11
8	1.0	8.00	9.43	8.97
8	0.7	2.90	6.79	4.52
8	0.5	1.67	4.67	2.90

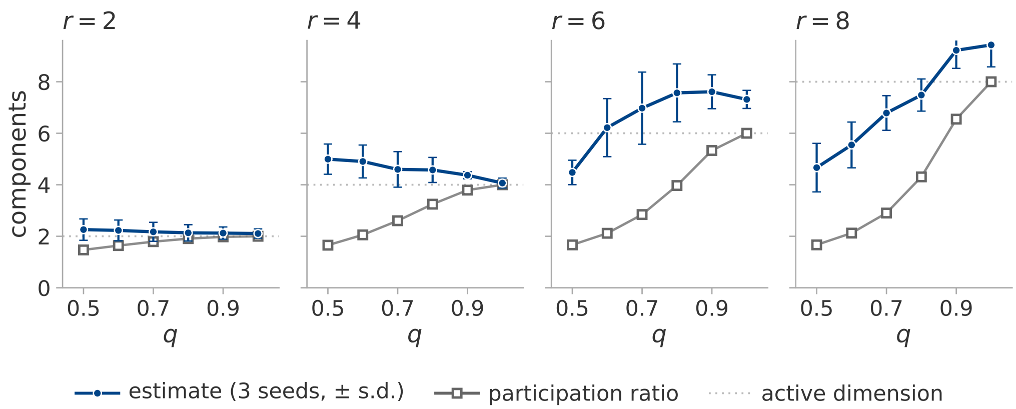

Figure 5: Anisotropy separates the two candidate estimands; the estimate follows the number of modes. The system of section 5.1 with mode l driven at amplitude q^l, three seeds, one panel per r. The manifold dimension is r in every panel, drawn dotted: the orbit closes onto an r-torus whatever q does. Filled circles are the estimate, the median over seeds, with their standard deviation as bars; open squares are the measured participation ratio of the covariance, which matches its closed form to two decimals and so has no spread.

At two and four directions the estimate is flat in q while the participation ratio declines, as intended. At six and eight it declines too. The reason is visible in the amplitudes: at $q = 0.5$ and $r = 8$ the weakest mode is driven at 0.008 of the strongest, far below the neighbourhood scale the estimator works at, so the torus it sees really does have fewer dimensions than the one that was built. The estimator therefore reports the number of modes it can resolve, which is neither the manifold dimension of the construction nor the participation ratio of its covariance, and coincides with both only when the amplitudes are equal.

F System constructions

Three constructions are described here: the image-data system of section 5.4, the function-space variant scored beside it, and the same system at twenty available directions.

The image-data system of section 5.4. The data is partitioned into twelve fixed groups, the same partition at every rank so that the partition is not itself a function of r; the loss weights of r of them are modulated by r rationally independent sinusoids filling one octave. Modulating group j tilts the gradient along a direction ϕ_j. These directions are neither orthogonal nor of equal effect, since random data groups have correlated gradients, so a mixing matrix $\mathrm{pinv}(\Phi)Q$ with Q an orthonormal basis of $\mathrm{range}(\Phi)$ orthogonalises them. The per-mode scalar response gain of the linearised update is then divided out. Without this correction the trajectory participation ratio falls well below r.

Table 15 verifies that each construction excites the rank it claims. Two of its rows are results rather than settings. A decaying transient does not recur whatever r is, since a converging trajectory is a curve; and plain mini-batch descent remains at 6.2 regardless of r, because its noise rank is fixed by the per-example gradient covariance of the data and is not a free parameter of the experiment. Projecting real mini-batch noise onto r directions recovers only part of the intended rank, which is why that regime is scored against its measured value and not against r.

Table 15: Whether each construction excites the rank it claims. Measured trajectory participation ratio by excitation regime and nominal rank, median over the four seeds, for the ten available directions of section 5.4. Every estimate in this paper is scored against these values, never against r.

regime	$r = 1$	2	4	6	8	what is excited
`qp`	1.00	2.00	4.00	6.00	8.00	r sinusoids, fast drive
`qp_slow`	1.00	2.00	4.00	5.99	7.98	the same at a training-log timescale
`qp_nopre`	1.00	2.00	3.73	5.98	7.77	`qp` without the preconditioner
`mixed`	1.00	2.00	4.00	6.00	7.99	`qp` plus noise in the same directions
`noise`	1.00	2.00	4.00	5.99	7.99	injected rank-r gradient noise
`noise_nopre`	1.09	1.98	3.71	5.24	6.75	the same without the preconditioner
`batch_proj`	1.00	1.74	3.08	4.19	5.38	real mini-batch noise projected to rank r
`batch`	6.58	6.58	6.56	6.60	6.56	plain mini-batch descent
`gd`	1.04	1.05	1.04	1.04	1.06	a decaying transient
`qp`, $\eta = 0$	1.00	1.00	1.00	1.00	1.00	requirement 4 check

In the last row the learning rate is multiplied by zero, so the active dimension is identically one at every nominal rank and an observer whose estimate still tracks r there is reading the drive rather than the optimiser.

Two properties of the backbone bear on the construction. The functional rank is 10 at every rank, seed and regime, with a participation ratio between 8.40 and 9.03; it belongs to the frozen backbone and does not depend on the excitation. The condition number of the drive-direction matrix is 1.00 at $r = 1$ by construction and has a median of 16.2 at $r = 8$, which is the anisotropy the mixing matrix exists to remove.

The function-space construction. A local adapter is built around a trained network of widths 64, 16 and 10. Its function-space Jacobian is QR-whitened, so that the first r columns have rank r at equal local scale. A quasiperiodic teacher excites every adapter coordinate and the student tracks it by exact backpropagation through both layers. The functional, trajectory-covariance and update-covariance ranks were all verified to equal r.

The twenty-direction system. The same construction at twenty available directions reproduces the nominal rank to within 0.008 components over $r = 1, \ldots, 20$ in the recurrent regime, and to three decimals up to $r = 10$. The other regimes behave as at ten directions: injected noise tracks r to within 0.29, projected mini-batch noise reaches only 9.8 at $r = 20$, and the decaying transient stays between 1.03 and 1.05 throughout.

G Diagnostics and figures for the grokking application

This appendix gives the diagnostic values behind section 7.1, a map of every training log in the plane of those diagnostics, and the label-matched pairs of section 7.1 beside the loss curves that account for them.

Table 16: Whether either grokking setting is admissible. The estimator and its diagnostics on training logs at $E_{\max} = 20$: the regularised stochastic setting through the parameter norm, the unregularised full-batch setting through the training loss. Every cell is a median over the run's sliding windows. Seven of the seventeen runs of table 24 are shown, and figure 6 plots all seventeen. Runs that have not generalised within the budget are censored, not negative.

run	generalises	MG	ρ_{ident}	roughness	$\mathrm{PR}_{\text{delay}}$	crossings
transformer, full data	yes	13.32	1.53	0.097	1.03	204
transformer, reduced data, s0	no	4.70	1.59	0.067	1.50	161
transformer, reduced data, s2	no	4.42	1.55	0.139	4.04	153
transformer, no weight decay	no	15.60	1.76	< 0.001	1.00	2
perceptron, $n + m$	yes	15.08	1.42	0.001	1.00	2
perceptron, $n \cdot m$	yes	14.97	1.39	0.001	1.00	2
perceptron, $n^3 + nm^2 + m$	no	15.41	1.51	< 0.001	1.00	2

Figure 6 places every training log of section 7 in the plane of the two diagnostics, against the region in which every accurate case of section 6 fell. No run of either setting is in it, and none is close: every one of the seventeen is unstable in the embedding dimension, the mini-batch runs at 1.40 to 1.59 and the full-batch and zero-weight-decay runs at 1.39 to 1.77, against the 1.10 the region allows. The two groups fail on the other axis in opposite directions, the full-batch series being monotone.

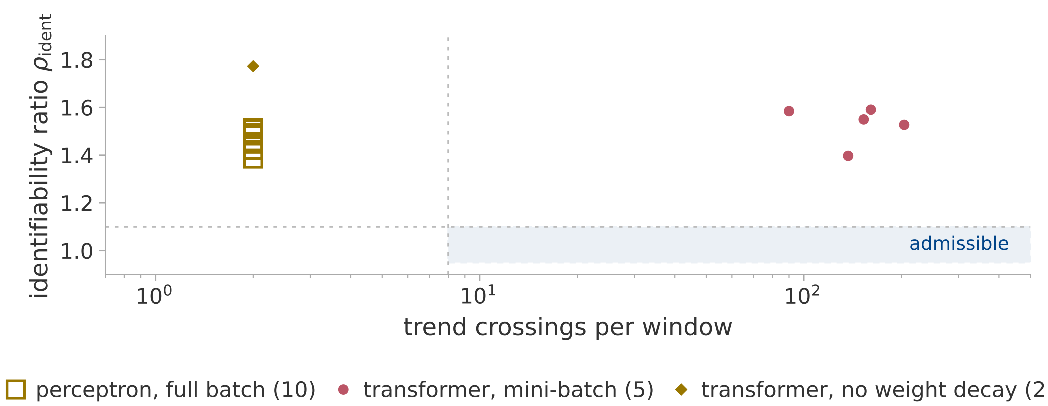

Figure 6: Whether any training log of either setting is admissible. Every log of section 7 in the plane of the two diagnostics, one point per run: seven transformer runs observed through the parameter norm, ten perceptron runs through the training loss. Colour is the regime of table 1 and marker shape the setting. The vertical axis is the identifiability ratio, near unity when the estimate does not depend on the embedding dimension; the horizontal axis is the trend-crossing count. The twelve full-batch runs share an abscissa of two crossings and are told apart by their ratios. The shaded rectangle is where every accurate case of section 6 fell. It describes the calibration and is not a decision rule.

Figure 7 shows the separation of section 7.1 and the mechanism behind it: the two members of a pair differ in the shape of their training curves long before any question of dimension arises.

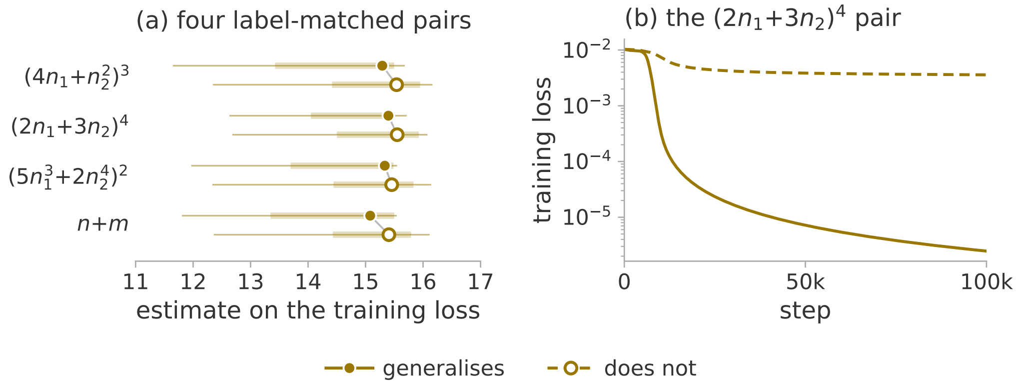

Figure 7: Whether the estimate separates label-matched pairs, and what accounts for the separation. (a) The estimate on the training loss for the four pairs of section 7.1. Both members of a pair reach full training accuracy under identical optimisation on identical data budgets. All eight runs are full-batch perceptrons, which figure 6 places in the transient regime, so all are drawn in one colour and generalisation is shown by fill and dash. The marker is the median over seven sliding windows, the thick bar their interquartile range and the thin line their full range. The medians separate the members of every pair with no overlap between the groups, but the window-level ranges overlap heavily, because the estimate drifts upward over training rather than scattering about a level. (b) The $(2n_1+3n_2)^4$ pair. The representable target drives the training loss three orders of magnitude below the perturbed one, though both fit the training set completely, so the two differ in shape rather than in any attractor dimension.

H The effective-rank collapse, run by run

Table 17 states the measurement of section 7.2 for every run. The plateau is the median of the 4 000 steps before the generalisation step, the dip the minimum within $\pm 4\,000$ of it, and "at" the position of that minimum relative to it. The two runs without weight decay have no generalisation step, and are measured in the window of the run they match.

The last column is the largest value the statistic reaches after its dip, which is what "re-expands" means: in parameter space it exceeds the plateau by an order of magnitude in every run, in function space it returns to about the plateau and exceeds it in three of the four.

Table 17: Plateau, dip and recovery per run, in function space and in parameter space. Depth is plateau divided by dip. The two rows without weight decay in each block are measured in the aligned window; their deepest fall anywhere in their own budget is 3.15 and 7.06 in function space and 1.66 and 2.37 in parameter space, which is the comparison section 7.2 also reports.

space	run	plateau	dip	depth	at	post-dip max
function	`mod_wd1`	24.24	3.80	6.38	+135	23.99
	`mod_wd1_s43`	33.02	3.86	8.55	455	27.80
	`mod_wd1_s44`	27.16	3.40	7.99	+515	30.82
	`s5_wd1`	21.04	2.62	8.03	1413	23.53
	`mod_wd0`	5.50	3.43	1.60	—	—
	`s5_wd0`	11.87	9.09	1.31	—	—
parameter	`mod_wd1`	3.01	1.09	2.76	+1 035	36.12
	`mod_wd1_s43`	2.57	1.15	2.24	1155	37.34
	`mod_wd1_s44`	2.71	1.13	2.39	+1 315	37.44
	`s5_wd1`	7.14	1.39	5.14	1513	12.46
	`mod_wd0`	29.93	27.70	1.08	—	—
	`s5_wd0`	5.57	4.49	1.24	—	—

I The trajectory sketch

The direct measurement of section 7.2 needs the trajectory itself, which is the whole parameter vector at each of two to three thousand logged steps. Every logged step is therefore compressed by a CountSketch (Charikar et al., 2002; Weinberger et al., 2009) to $\mathbb{R}^{1024}$, with two independent hash families. Two spaces are recorded: the parameters, and the probe logits after centring each example over classes and normalising to unit length. The normalisation is necessary: the raw logit scale is mechanically coupled to weight decay, so a signal read off it could not be distinguished from a measurement of weight decay.

We do not verify a spectral guarantee for the compression. The elementary bound for a sketch of this kind is on pairwise inner products (Weinberger et al., 2009), and a participation ratio is a functional of the whole spectrum, on which an entrywise Gram bound is weak. A subspace-embedding guarantee does exist for CountSketch (Clarkson & Woodruff, 2013) and would apply here, since a sixty-sample window has rank at most sixty, but it requires a rank we do not measure per window. The two hash families supply a direct estimate of the error instead: every reported value is the mean of the two, with their disagreement recorded beside it. Over the published transformer windows that disagreement is $1.1\,\%$ of the value at the median and $8.4\,\%$ at its worst, which is the number the reader should use; it is smaller than every effect section 7.2 reads.

Three checks stand behind the numbers. *The sketch is accurate on synthetic trajectories of known rank.* Table 18 passes such trajectories through it and compares against the same statistic computed on the uncompressed trajectory: the sketch is accurate to 0.04 components throughout. The shortfall against the nominal rank at 5 and 10 is present without any compression, so it belongs to the participation ratio on a sixty-sample cloud and not to the sketch. Nothing above rank ten is tested, however; the plateaus in figure 2 are higher than that.

It agrees with exact arithmetic on a real run. On the perceptron architecture, where the parameter count is small enough to allow it, the participation ratio computed on all $145\,500$ raw parameters agrees with the sketched value to three decimals.

The observer does not perturb training. The log is required to be bit-identical with and without it, over four hundred steps at reduced probe size. That check matters because the task module seeds one global random stream that the train–validation split, the initialisation and the mini-batch order all continue: a single stray draw changes the initial weights and can destroy grokking.

Table 18: What the compression costs. The sketch against the same statistic computed without it, on synthetic trajectories of 60 samples and known rank in $145\,500$ dimensions. The last column is the price of the compression; the difference between the first two is the price of a sixty-sample participation ratio, and is not attributable to the sketch.

true rank	PR, uncompressed	PR, sketched	sketch − uncompressed
1	1.00	1.00	0.00
2	1.99	1.98	−0.01
5	4.42	4.46	0.04
10	8.61	8.63	0.02

Windows are sixty logged rows, which is 600 optimiser steps for the modular runs and 300 for S_5, and are labelled by their centre, so a feature at a given step is located to within half a window and the window is not causal. Five statistics are computed per window in each space, of which the paper reports two: $\mathrm{PR}^{\mathrm{det}}$, the participation ratio of the positions after removing a per-window linear trend, since a steady drift is rank one and would otherwise mask everything else; and $\mathrm{PR}^{\mathrm{upd}}$ on the increments block-averaged over five logged steps, to suppress the mini-batch noise floor. Beside them the total displacement is recorded, as the null for a rank that falls simply because the trajectory stopped moving.

J THE FULL-BATCH SETTING

Repeating the direct measurement on the perceptron setting gives a result that must be read as a limit of the measurement rather than as a negative. Under full-batch gradient descent every window of every run has a participation ratio between 1.000 and 1.362 on the four statistics the comparison uses, and between 1.000 and 2.171 over all eight recorded columns. This is not an artefact of the sketch, which appendix I checks against exact arithmetic on the same architecture.

A deterministic update makes the trajectory a smooth curve. Over a 600-step window such a curve is straight to within the measurement precision, so the statistic reports the curvature of the trajectory at the scale the window sets, and at that scale there is almost none. The statistic has no range in which a collapse could appear, so the absence of a feature is not evidence of its absence: the window must be set in units of the curvature scale of the trajectory.

Repeating the measurement against window length confirms the diagnosis and closes the opening it appeared to leave. Two runs of $150\,000$ steps, a generalising one and its label-matched partner, are measured over windows spanning 600 to $120\,000$ optimiser steps, with the window lengthened by spreading sixty samples further apart rather than by adding samples, so that the bound on the statistic and its noise floor stay exactly where the published measurement put them and only the span changes. Table 19 gives the result. The statistic does leave the floor, at a scale of about 10^4 steps, two orders of magnitude above the window that produced the 1.000–1.362: the absence of a feature there was a resolution limit and not a fact about the run.

It is not, however, a hidden signal. At $12\,000$ steps one localised maximum appears, of the same size in both runs (table 19); function space agrees, with more range and no more separation, 2.618 against 2.145. Nor can the maximum be placed relative to either milestone: the two centres are adjacent points of a $3\,450$-step grid and their windows overlap over $6\,900$ to $15\,250$ steps. Measured against memorisation the two offsets are $+3\,600$ and -450, a difference this design cannot resolve — a $12\,000$-step window cannot place a feature to better than the interval between memorisation and generalisation.

The claim the experiment does support is the negative one: at the window where the statistic has range, the generalising run and its label-matched control are not separated by it. Beyond $30\,000$ steps the ordering reverses and the non-generalising run moves further, because the window is then a large fraction of a run whose weight norm is still climbing and the statistic is tracking that drift. The full-batch trajectory therefore carries no rank signature of generalisation at any window length between 600 and $118\,000$ steps: the collapse of section 7.2 belongs to the regularised stochastic setting, and a trajectory participation ratio must be controlled for window length as well as for batch size before anything is concluded from it.

Table 19: Whether the full-batch statistic has any range at longer windows. Detrended position participation ratio in parameter space against window length, full-batch gradient descent at $p = 97$, over all placements of the window. Sixty samples per window at every length; the lengths are quarter-multiples of the generalisation step of the run that has one. The 600-step row is the measurement reported above, on the same configuration and seed.

	generalises		control, never generalises	
window (steps)	median	max	median	max
600	1.002	1.035	1.001	1.011
2 950	1.066	1.258	1.003	1.087
5 900	1.027	1.534	1.005	1.360
11 800	1.010	1.951	1.020	1.826
29 500	1.006	2.047	1.112	1.735
59 000	1.025	1.551	1.345	1.912
118 000	1.231	1.626	2.009	2.271

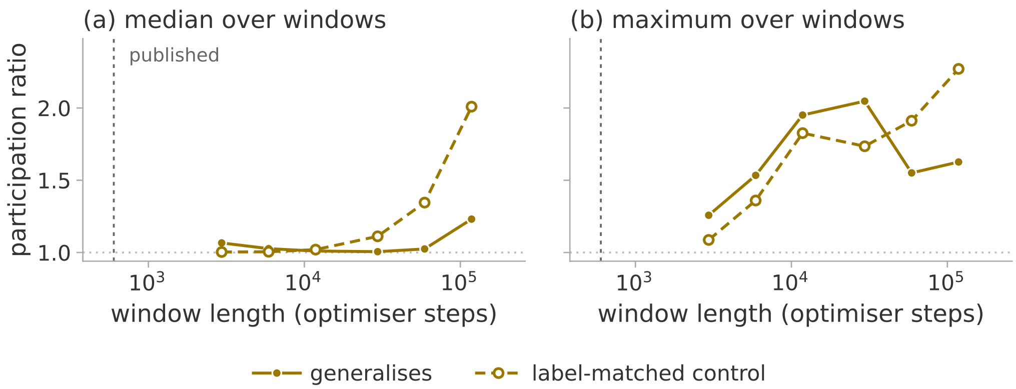

Figure 8: The trajectory participation ratio against window length, on the full-batch pair of this appendix: modular addition and its label-matched partner, which memorises but never generalises. The statistic is $\mathrm{PR}^{\mathrm{det}}$ on the sketched parameter trajectory over windows of $3\,000$ to $120\,000$ optimiser steps, lengthened by spreading sixty samples further apart rather than by adding samples, so that only the span changes. (a) The median over a run's windows, (b) the maximum; the dotted line at 1 is the floor a straight trajectory returns, and the dashed vertical the 600-step window of section 7.1. Both runs are transient, so both are drawn in one colour and the non-generalising one is open and dashed.

Introducing mini-batches confirms the diagnosis. The same statistic then ranges over more than a decade while the generalisation step moves by one logged step, so the learning is unchanged and the statistic has moved by an order of magnitude. Any claim based on a trajectory participation ratio must therefore be controlled for batch size and noise scale before it is attributed to learning.

K Declines of the estimate that are not generalisation

A decline of the estimate is not specific to generalisation. In a $200\,000$-step run at $p = 211$ with no weight decay, which memorises at step $1\,800$ and ends at $3.6\,\%$ validation accuracy, a decline is detected $49\,250$ steps after memorisation. The ramped-gain experiment of section 6.3 supplies a mechanism that needs no dimension at all: a ramped observer gain moves the estimate by 1.53 components, while this run's parameter norm grows from 43 to 126 over the budget. It is the growth section 7.2 reports in the two runs without weight decay, acting through the observer.

That run does not show a simplified trajectory. Its diagnostics place it in the transient regime (table 16), where the estimate is not a dimension, so a decline in it is a decline of a quantity we have argued should not be read as one. The two must be kept apart. The direct measurement of section 7.2 is an effective rank of the trajectory itself, and it declines in four generalising runs and not in the two without weight decay. The estimate on a log, at the frozen window and in these settings, is neither a dimension nor an effective rank, and it declines in at least one run that does not generalise. The defensible claim is therefore the weaker one: a decline on a log is not evidence of generalisation.

Section 7.3 does not overturn it. There the log estimate declines where the trajectory does, in four runs and in neither of the other two, but at a window two orders of magnitude shorter than this one and under a configuration chosen for that window; the level is no more interpretable there than here, and only a change against the run's own distribution of changes is claimed. Nor is the absence of generalisation within a budget evidence that a decline was a false alarm: two reduced-data configurations against which earlier specificity figures were computed generalise once the budget is extended, so such runs are censored and no specificity figure may be computed against them.

L Resolution of a windowed estimate on a training log

The claim of section 7.1 that a change within a run could be read against a fixed configuration is testable on the seven extended runs. The test returns a resolution limit rather than an answer. The frozen configuration takes a window of a third of each record: $4\,000$ logged samples, which at the logging stride is $39\,990$ optimiser steps, advanced by $10\,000$ steps. That is nine windows per run and sixty-three weight-norm windows in all, localising a feature only to $\pm 19\,995$ steps, sixty-seven times coarser than the direct measurement of section 7.2.

The consequence is that the alignment figure 2 performs cannot be performed here. Of the twenty-seven window centres belonging to the three generalising runs, one lies within $5\,000$ steps of its own generalisation step and two within $10\,000$; no two runs share an offset, so no offset has more than one run defined and no pooled mean exists. For the fastest-grokking run the entire pre-generalisation phase is thirteen per cent of one window; for the reduced-data run that generalises at $110\,940$, no window centre falls after the transition at all.

The runs are consistent with that. The one run with a centre near its generalisation step falls by 3.54 components over the next stride, three times the floor. That step, however, lies inside four of its nine windows, and the falling windows are those in which its parameter norm is itself moving fastest, which is the ramped gain of section 6.3. The other two runs show nothing.

Nor is any of this specific. Across the fifty-six stride-to-stride changes, six of twenty-four clear the floor in the runs that generalise and six of thirty-two in the runs that do not. The largest change in the set, 6.36 components, is in a run that does generalise, but it falls nearly ninety thousand steps after its generalisation step; the next largest, 5.20, is in a run that ends at seven per cent validation accuracy. The tightest comparison available is three runs of one configuration, of which one generalises: their ranges are 0.77, 0.78 and 0.67 components, with nothing to separate them.

This repeats the argument of appendix J. At the window the configuration provides, the statistic has no range in which a feature localised to the transition could appear, so its absence is not evidence of absence. Testing the claim needs a window set in units of the width of the transition, which for these runs is hundreds of steps and not tens of thousands: a re-run of the estimator, not a re-reading of what we have.

The matched-window re-run. Section 7.3 performs it, on the six runs of figure 2 rather than the seven of this appendix, because those are the runs whose trajectory was stored and only a paired comparison answers the question. The window is 60 samples throughout, which is 600 optimiser steps for the modular runs at their stride-10 logging and 300 for the S_5 runs at stride 5, centred on the midpoints of the windows of section 7.2; since those are 600 and 295 steps wide, every log window matches the one it is paired with to within two per cent. Four things qualify the result.

The configuration is not the frozen one, and cannot be. At 60 samples the frozen delay span, $(20-1)\times 4 = 76$, exceeds the window. The grid crosses $E_{\max} \in \{4, 6, 10\}$, $\tau \in \{1, 2, 4\}$, $k \in \{5, 20\}$ and both Theiler rules, thirty-six cells, of which twenty-six are long enough to return a value. Twenty-four of those separate the four generalising runs from the other two; the two exceptions are $E_{\max} = 4$ and $E_{\max} = 6$ at $\tau = 1$, $k = 5$ under the embedding rule. The headline cell is named by the rule of section 7.3 and is not the best cell: $E_{\max} = 6$ at the same lag, neighbour count and Theiler rule gives a shallowest generalising fall of 2.44 against its 1.98.

The admissibility diagnostics do not survive the shortening. The identifiability ratio needs $2E_{\max}$. At $E_{\max} = 10$ the doubled delay span does fit the window, but the Theiler exclusion then leaves two usable points against $k = 20$, so the headline cell has no ratio. Where it can be computed, at $E_{\max} \le 6$, it reads 0.97 to 1.00 under the frozen Theiler rule and 1.29 to 1.62 under the other, and settles nothing. That is why section 7.3 reads a change and not a level.

Linear detrending removes the effect, and the surrogate test does not. Repeating every cell on the linearly detrended window — the removal $\mathrm{PR}^{\mathrm{det}}$ performs per coordinate — leaves no cell separating the two groups. The two removals are not comparable: on a 1024-dimensional sketch a per-coordinate linear detrend takes one direction of many, while on a scalar it takes most of the series. The surrogate test keeps the shape and destroys only the fine structure, which is why it is the one that discriminates. Both are reported in table 20.

The observer matters. The result above is the parameter norm. On the training loss ten of the twenty-six cells separate the groups and the surrogate test passes in the S_5 run only; on the validation loss fifteen do. That observer defines the event being aligned on, so it is listed for completeness and not used.

The two runs without weight decay have almost no dynamic range. At this window the baseline belongs to the configuration and not to the run: on a standardised 60-sample straight line equation (3) returns 2.7733, within 0.28 of every run's pre-transition level. The modular one's parameter-norm log is straight to $0.7\,\%$ of its own spread across the whole comparison interval, so its estimate cannot move — a span of 0.0029 components over fifty windows, against 2.05 to 2.56 in the generalising runs. Its $D = 1.00$ therefore records that the observer offered the estimator no local structure to respond to, not that a dimension was measured and found unchanged: a statistic with three thousandths of a component of range could not have shown a two-fold fall whatever the trajectory did. The S_5 run is a weaker case of the same thing, $3.2\,\%$ residual and a 0.22-component span.

This forbids treating their D and quantile columns as informative dimensional negatives. It leaves the surrogate test untouched, since there each run is compared only against surrogates of itself, and it leaves the four generalising runs untouched. It also reinforces the point already made: the separation between the groups is mechanically a separation in local roughness.

Table 20: Whether the log estimate falls at generalisation, and whether shape-preserving surrogates reproduce that fall. The matched-window re-run of section 7.3 on the parameter-norm log, at the headline cell $E_{\max} = 10$, $\tau = 1$, $k = 20$, embedding-span Theiler. D is the median estimate over $[t_{\text{gen}} - 3000, t_{\text{gen}} - 1000]$ divided by the minimum over $[t_{\text{gen}} - 1000, t_{\text{gen}} + 2000]$; the quantile is its rank among the same statistic at every window centre after memorisation; the floor is where the minimum falls. The p columns are the fraction of 39 surrogates, at each smoothing length in samples, whose D is at least the observed one, as a median over five seedings, which is why a median is reported at all. The runs without weight decay are aligned on the generalisation step of their matched run.

run	D	quantile	floor	p_{101}	p_{201}	p_{401}
modular, seed 42	2.96	0.99	-65	0.025	0.025	0.025
modular, seed 43	1.98	0.93	255	0.050	0.050	0.025
modular, seed 44	2.14	0.92	$+315$	0.050	0.075	0.025
S_5 composition	2.12	0.89	1313	0.050	0.050	0.050
modular, no decay	1.00	0.46	-865	0.950	0.950	0.925
S_5, no decay	1.04	0.38	1613	0.975	0.975	1.000

The same comparison on the two nulls reported beside the estimate: $\mathrm{PR}_{\text{delay}}$ gives 1.01 to 1.49 on the four generalising runs against 1.00 on the other two, and the roughness gives 6.5 to 50.8 against 1.41 on both. The linear participation ratio therefore responds less than the estimator and the roughness far more, which is what section 7.3 reports.

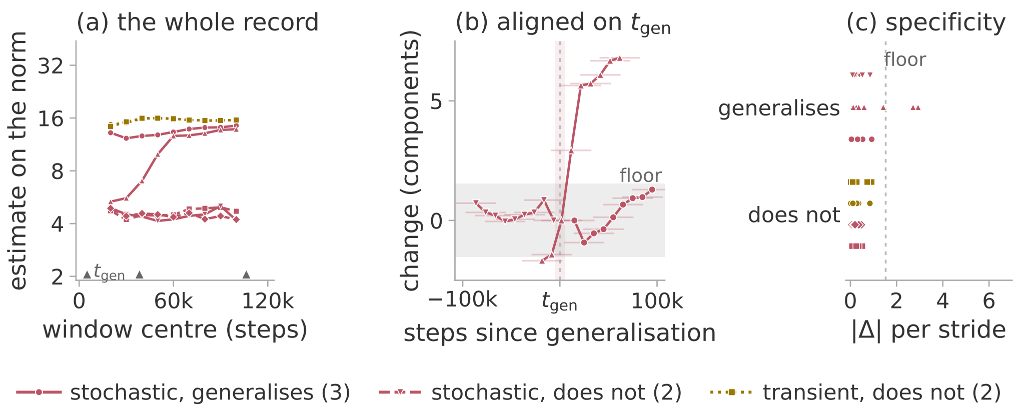

Figure 9: What a frozen-window estimate can and cannot localise. The windowed estimate on the parameter norm for the seven extended runs, under the frozen configuration of section 7.1: nine windows per run of 39 990 optimiser steps, advanced by 10 000, plotted at their centres. Colour is the regime of table 1. (a) All sixty-three windows, with the three generalisation steps marked on the axis. (b) The three generalising runs aligned on their own generalisation step, each point a change from the window before it and its bar the span it summarises, so nothing is localised to better than $\pm 20\,000$ steps. The grey band is the ± 1.53-component floor of section 6.3, the shaded column $\pm 5\,000$ steps of t_{gen}. (c) Every stride-to-stride change, one sub-row per run, against the same floor.

M THE REPRESENTATION DIMENSION IN CLOSED FORM

The perceptron setting of section 7 makes the learned representation analytically available. Three different numbers can be read off it. They are not commensurable, and the literature does not always separate them.

The *mode count* is $(p+1)/2$, or 49 at $p = 97$: the number of distinct Fourier frequencies the generalising solution of Gromov (2023) uses. The *order parameter* is the mean inverse participation ratio of the first layer's Fourier spectrum, which is 1.000 for that solution because each neuron carries a single frequency, and about 0.041 at random initialisation; it measures how concentrated each neuron is, not how many modes exist. The *effective rank* is the participation ratio of the singular values of the first-layer weight matrix, which for the analytic solution is 148.8: that matrix is 500×194 and full rank, because each non-zero frequency contributes a sine and a cosine direction in each of the two input blocks, so 148.8 measures how evenly those 194 directions are used rather than counting them. Only the first is a dimension in the sense the closed form fixes. Two things follow from measuring the other two, and both qualify readings of the representation dimension elsewhere.

The representation is not complete at the transition. Table 21 follows the order parameter through a modular addition run. It does not jump at the transition: it begins rising at memorisation, passes smoothly through, and goes on rising for tens of thousands of steps afterwards without reaching the analytic value. Generalisation therefore arrives well before the representation is complete, and requires it only to be periodic enough.

Table 21: Whether the representation is complete when generalisation occurs. The order parameter through one modular addition run at $p = 97$, full-batch gradient descent with no weight decay, against the analytic solution of Gromov (2023). The run generalises at step 11 950; the budget ends some 87 000 steps after validation accuracy first reaches 100 %.

point in the run	order parameter
just after memorisation	0.067
just before generalisation	0.189
validation accuracy first reaches 100 %	0.263
end of the budget	0.595
the analytic generalising solution	1.000

The effective rank of the same layer carries almost no signal: it reads 139.1 at random initialisation, within seven per cent of the analytic 148.8 before any training, then spans 132.6 to 150.9 over the run without being monotone and overshoots the reference before falling back. Agreement with the reference is therefore not evidence of anything.

The order parameter is only interpretable against its own reference. Table 22 gives the measured value beside each task's own closed form. Two of the tasks that generalise have a reference near the floor, because the representation is periodic in $g_1(n)$ while the spectrum is taken over the raw index n, and g_1 is not linear there. For such a task a measured 0.044 lies near its own reference of 0.062 and indicates convergence; against the 1.000 that modular addition sets, the same value would indicate that nothing had been learned. The tasks that never generalise remain at the initialisation floor. The analysis returns no reference for them rather than substituting one from another task.

Table 22: Fourier inverse participation ratio of the first layer at the end of training, against each task's own closed form. All runs are full-batch gradient descent with no weight decay at $p = 97$. A dash marks a task for which the source papers give no closed form; the four tasks below the rule, whose grokking step reads "none", are the ones that never generalise.

task	measured	own reference	grokking step
$n + m$	0.595	1.000	11 760
$n - m$	0.582	1.000	11 540
$n^2 + m^2$	0.093	0.052	10 470
$n \cdot m$	0.073	—	11 560
$(4n_1 + n_2^2)^3$	0.243	1.000	12 290
$(2n_1 + 3n_2)^4$	0.283	1.000	7 680
$(5n_1^3 + 2n_2^4)^2$	0.044	0.062	6 600
$(4n_1 + n_2^2)^3 + n_1 n_2$	0.040	—	none
$(2n_1 + 3n_2)^4 - n_1^2$	0.043	—	none
$(5n_1^3 + 2n_2^4)^2 - n_2$	0.039	—	none
$n^3 + nm^2 + m$	0.040	—	none

N THE EXCLUSION CAP AT THE TWENTY-DIRECTION CONFIGURATION

The Theiler exclusion is meant to remove neighbours that are close in time rather than in phase space. The rule this study asks for is $W_T = \max((E-1)\tau,\ t_{\text{acf}})$: at least the full span of a delay vector, so that two neighbours cannot be near each other merely because their windows overlap. The implementation caps W_T at 150 samples.

At the eight-direction configuration the embedding span is $(20 - 1) \times 4 = 76$, so the cap binds only where the autocorrelation time exceeds 150: never on the calibration systems, whose autocorrelation time is one to three samples, but on five of the seven parameter-norm records of section 7, where it is 161 to 858. Those five were scored at $W_T = 150$ rather than at the value the rule asks for, in the direction that inflates the estimate. At the twenty-direction configuration the span is $(40 - 1) \times 16 = 624$, so the cap binds unconditionally and every row of those scores records

$W_T = 150$, under a quarter of what the rule asks for. This is a departure from the stated protocol rather than a deliberate choice. It is why section 5.4 marks that result exploratory.

Table 23 measures what it costs, by re-scoring the stored trajectories at both exclusions. Nothing is regenerated, so the systems cannot drift between the two settings.

Table 23: What the exclusion cap costs. The twenty-direction system re-scored at the capped exclusion and at the one the configuration asks for. Seven ranks, three seeds, the three observers with the lowest error, median over seeds; the error is against the measured $\mathrm{PR}^{\mathrm{pos}}$ and is not comparable with table 5, which covers more observers. Only the difference between the columns is meaningful.

r	measured $\mathrm{PR}^{\mathrm{pos}}$	MG at $W_T = 150$	MG at $W_T = 624$
2	2.00	1.74	1.81
5	5.00	4.16	5.15
8	8.00	6.28	6.71
11	11.00	9.54	10.26
14	14.00	9.81	9.37
17	17.00	14.40	14.19
20	19.99	15.92	16.60
mean absolute error		2.16	1.89
Spearman ρ		1.00	0.96

Three things follow. The ordering, the claim section 5.4 makes at this range, is nearly unaffected: the rank correlation is 1.00 at the capped exclusion and 0.96 at the one the configuration asks for, a single inversion between $r = 11$ and $r = 14$. The mean absolute error falls from 2.16 to 1.89 components, so enforcing the protocol makes the recovery slightly better rather than worse, and the cap was not flattering the reported result. The saturation above about six directions is present at both settings and is therefore not an artefact of the exclusion.

The same cap has a second effect that the identifiability ratio inherits. The exclusion actually applied is $W_T = \min\big(\max((E-1)\tau,\, t_{\mathrm{acf}}),\, 150\big)$, and the ratio's numerator is computed at $2E_{\max}$ with τ held fixed. At the eight-direction configuration the numerator's span is therefore $(40 - 1) \times 4 = 156$ and the denominator's 76: the numerator is capped at 150 whatever the data does, while the denominator is capped only once t_{acf} reaches 150. The two are treated identically exactly when $t_{\mathrm{acf}} \geq 150$ and differently when it is smaller.

Five of the seven parameter-norm records of section 7 are unaffected by this. The asymmetric cases are the other two and, more consequentially, the calibration systems of section 5, whose autocorrelation time is one to three samples, so that the reference band of section 6.4 was itself measured with 76 against 150. The effect is small at this span and the diagnostic separates the regimes of section 6 without overlap, but a ratio built from two differently excluded estimates is not the quantity section 3.4 describes, and lifting the cap is the first thing to change in any follow-up.

O Run inventory

Table 24 lists every training run the paper draws on. Two architectures appear: **T** is the one-layer transformer of Nanda et al. (2023), $d_{\mathrm{model}} = 128$, four heads, no layer normalisation, 226 816 parameters on modular addition at $p = 113$ and 228 608 on S_5, trained in double precision with AdamW at learning rate 10^{-3} and $\beta = (0.9, 0.98)$; **P** is the two-layer quadratic perceptron of Gromov (2023), 145 500 parameters at $p = 97$ and width 500, trained by full-batch gradient descent with no regularisation.

Three departures from the source configurations should be stated, since two of them place the runs in the regimes section 7.1 assigns. Nanda et al. (2023) train full batch; we use mini-batches of 256, which makes this setting stochastic rather than deterministic. The S_5 weight decay of 0.2 and the S_5 task itself (Power et al., 2022) are outside that configuration. The two S_5 runs also step the optimiser twice per batch, reproducing a duplicated update in our own earlier released logs, so their effective learning rate and decay are about double the nominal ones.

Definition of memorisation and generalisation. Memorisation and generalisation are the first logged step at which the training and the validation accuracy respectively reach 95 %, with no persistence requirement. Training accuracy is measured on the current mini-batch and validation accuracy on a fixed 512-example subset of the validation split, so both carry a sampling error of a few tenths of a per cent; the step is resolved only to the logging stride, which is 10 steps for the modular runs and 5 for S_5. The extended reruns of section 7.1 instead require the threshold to hold over a sustained block, which is the stricter rule; the two rules put memorisation within 70 steps of each other on every run here and differ by 1 080 steps on one generalisation step. No claim turns on either.

A dash means the criterion was not met inside the budget. Nothing here establishes that such a run would never generalise, and table 24 records the final validation accuracy so that the reader can see how far from the threshold each one ended.

Table 24: Training runs used in this paper. α is the training fraction, "final" the validation accuracy at the end of the budget. A dash in the generalisation column marks a run that had not generalised when its budget ended; such runs are censored observations, not established negatives. The name `s5_wd1` denotes the weight-decay member of the S_5 pair, whose decay is 0.2; it is not a decay of 1.

run	task	arch	wd	α	budget	mem.	gen.	final	used in
Trajectory sketch attached									
`mod_wd1`	mod. add. p=113	T	1.0	0.30	20k	1 470	13660	1.00	section 7.2
`mod_wd1_s43`	mod. add. p=113	T	1.0	0.30	20k	1 580	6640	1.00	section 7.2
`mod_wd1_s44`	mod. add. p=113	T	1.0	0.30	20k	1 410	3 680	1.00	section 7.2
`s5_wd1`	S_5 composition	T	0.2	0.50	15k	1 280	6685	1.00	section 7.2
`mod_wd0`	mod. add. p=113	T	0.0	0.30	20k	630	—	0.03	section 7.2
`s5_wd0`	S_5 composition	T	0.0	0.50	15k	1 015	—	0.01	section 7.2
Extended reruns, 120 000 *steps*									
`grokpos_s0`	mod. add. p=113	T	1.0	0.30	120k	1 450	5070	1.00	section 7.1
`lowdata15_s0`	mod. add. p=113	T	1.0	0.15	120k	620	—	0.58	section 7.1
`lowdata15_s1`	mod. add. p=113	T	1.0	0.15	120k	630	106 380	1.00	section 7.1
`lowdata15_s2`	mod. add. p=113	T	1.0	0.15	120k	660	—	0.01	section 7.1
`lowdata20_s0`	mod. add. p=113	T	1.0	0.20	120k	800	38380	1.00	section 7.1
`wd0_s0`	mod. add. p=113	T	0.0	0.30	120k	610	—	0.09	section 7.1
`wd0_s1`	mod. add. p=113	T	0.0	0.30	120k	630	—	0.07	section 7.1
`p211_wd0`	mod. add. p=211	T	0.0	0.16	200k	1 800	—	0.04	section 7.1
Perceptron, $p = 97$, *full batch, no regularisation*									
`a_add`	$n + m$	P	0	0.5	100k	9230	11760	1.00	section 7.1
`a_mul`	$n \cdot m$	P	0	0.5	100k	9100	11560	1.00	section 7.1
`a_sub`	$n - m$	P	0	0.5	100k	9110	11540	1.00	appendix M
`a_sq_sum`	$n^2 + m^2$	P	0	0.5	100k	8440	10470	1.00	appendix M
`a_sum_sq`	$(n + m)^2$	P	0	0.5	46k	7350	8190	1.00	appendix M
`x_mix_quad`	mixed quadratic	P	0	0.5	100k	9600	—	0.01	figure 6
`x_no_grok`	$n^3 + nm^2 + m$	P	0	0.5	100k	9860	—	0.01	section 7.1
`g_p1`	$(4n_1 + n_2^2)^3$	P	0	0.5	100k	7470	12290	1.00	section 7.1
`g_p1x`	$+ n_1 n_2$	P	0	0.5	100k	10140	—	0.01	section 7.1
`g_p2`	$(2n_1 + 3n_2)^4$	P	0	0.5	100k	6470	7680	1.00	section 7.1
`g_p2x`	$- n_1^2$	P	0	0.5	100k	9730	—	0.02	section 7.1
`g_p3`	$(5n_1^3 + 2n_2^4)^2$	P	0	0.5	100k	5970	6600	1.00	section 7.1
`g_p3x`	$- n_2$	P	0	0.5	100k	9420	—	0.74	section 7.1

Four of the perceptron runs are repeated with the trajectory sketch attached, for the measurement of appendix J: `a_add` and `x_no_grok` from the table above, together with the label-matched pair `g_p1` and `g_p1x`. Each is run at full batch and again with mini-batches of 512, and two of them once more at $p = 23$. The eight at $p = 97$ agree with the rows above on their memorisation and generalisation steps to within one or two logged steps, so the comparison of appendix J is one of noise rather than of learning.

Two notes on the censored rows. `lowdata15_s0` ends its budget at 58 % validation accuracy and still rising, so it is the clearest censored observation in the set rather than a negative. Its sibling `lowdata15_s1` generalises at 106 380, five sixths of the way through the same budget: at this training fraction the two seeds differ by more than the budget can resolve, which is the point section 7.1

makes of them. Finally, `g_p3x` reaches 73 % validation accuracy against a majority-class baseline of 1.3 % while being provably outside the class the architecture can represent, so it is the one row in this table that neither of the source papers accounts for.

P THE THEILER EXCLUSION AND THE DIMENSION OF A TRANSIENT

Section 3.3 claims that recurrence is a requirement of this protocol and not of the estimator. The claim is testable on the excitation cases of section 5.4 by moving one setting and nothing else: the Theiler exclusion W_T, from zero to the frozen rule. Four observers, one per family, three seeds, $r \in \{2, 4, 6, 8\}$, seven windows each; the estimator is otherwise the frozen configuration, and at the frozen setting every cell reproduces section 5.4 to every stored digit. Table 25 gives the result.

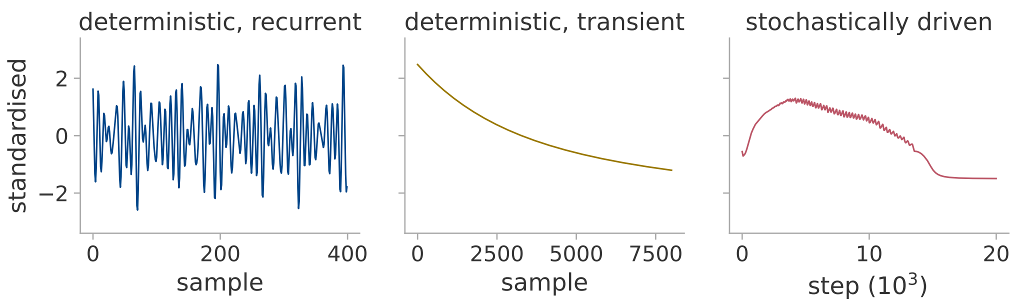

Figure 10: What the estimator is given in each regime. The three regimes of table 1 as raw scalar logs, standardised as the estimator standardises them. Left, the recurrent case of this appendix, whose orbit closes onto a torus. Middle, the transient case over its whole record, a decay that increases on none of its eight thousand steps. Right, a mini-batch transformer run through the parameter norm. The last two are the ones the diagnostics of section 3.4 must reject, and the difference between them is why one statistic will not do it: the transient is monotone and the driven run is not, so a test of non-monotonicity catches the first and a test of embedding stability the second.

Table 25: The estimate against the Theiler exclusion, median over observers, seeds and windows. The transient's estimand is 1: at $W_T = 0$ the estimator returns it, and the exclusion removes it. The recurrent cases are unchanged under the same variation, and 28.96 is not a level but the value a divergent quantity happens to take at the cap of 150 the implementation imposes.

W_T	0	2	5	20	100	150	uncapped
recurrent, fast	5.39	5.42	5.42	5.42	5.42	5.41	5.41
recurrent, slow	2.11	2.92	3.23	3.32	3.33	3.33	3.33
transient	1.20	1.71	2.38	5.51	20.83	28.96	39–172

Three things follow. *The number at $W_T = 0$ is the right one.* On a uniformly sampled curve the 20 nearest neighbours lie at offsets $\pm 1 \cdots \pm 10$, for which equation (3) returns 1.19928 exactly; the transient case returns 1.1971 ± 0.0010 across all forty-eight cells. The excess over unity is the lattice bias of a deterministically sampled curve, not an error; the measured trajectory participation ratio of the same case is 1.02 to 1.11.

The mechanism is the loss of near neighbours, not of neighbours. The estimator fills its twenty slots at every exclusion; only their distance changes. Before any exclusion the median temporal separation of a point's twenty nearest neighbours is 2546 samples on the fast case and 6 on the transient, where the nearest neighbours in space simply *are* the temporal ones. Excluding 150 samples costs the fast case almost nothing, 138 048 close pairs surviving of 138 761, and costs the transient all 131 849 of its own. Its twenty retained distances then span a factor of 1.07, their log-ratio sum falls from 15.86 to 0.63, and the reciprocal in equation (3) diverges. No cell is marked degenerate, so this is not a numerical floor. An exponential decay with no optimiser and no data reproduces both ends, 1.196 and 26.4.

Two qualifications. Recurrence is not always needed even here: in the fast case $W_T = 0$ already returns r, because it is not oversampled. The exclusion rescues the slow case instead, whose nearest neighbours lie 574 samples away, from a rank-blind 2.1 to its published 3.5. A transient also need not have dimension one: a decaying *oscillation* returns 2.36 at every exclusion, correctly, because at $k = 20$ the neighbourhood spans several turns of the spiral and the two-dimensional sheet they lie on is all there is to see at that resolution. The asymmetry is general; the value is not.

A recurrence-based diagnostic. $\hat{d}_{MG}(W_T{=}150)/\hat{d}_{MG}(W_T{=}20)$ is 1.000 on the fast case and 1.001 on the slow, against 5.25 on the transient, with no overlap over forty-eight cells per case (largest recurrent 1.012, smallest transient 4.71); the less expensive $\hat{d}_{MG}(100)/\hat{d}_{MG}(50)$ separates as cleanly. Unlike the trend-crossing count it tests recurrence in the reconstructed space rather than non-monotonicity of the series, which is the gap section 3.4 leaves open and which appendix Q shows to matter on real runs. We report it rather than adopt it: it is calibrated on the same systems that produced the two diagnostics and has not been validated out of sample. The principled alternative is older than either. A space–time separation plot (Provenzale et al., 1992) shows directly how neighbour distance depends on temporal separation, and the variation above is a coarse version of one. Choosing the exclusion by such a measurement rather than by rule is the obvious improvement to the protocol.

Q Descent at the edge of stability

Every system of section 5 is driven, so that evaluation establishes a conditional, and section 6.1 asks whether anything undriven occupies the admissible regime. The one candidate the literature offers is the edge of stability (Cohen et al., 2021): gradient descent at a fixed rate sharpens until the top Hessian eigenvalue reaches $2/\eta$ and then stays there, while the loss descends on long scales and not on short ones. That is deterministic and not a plain transient, which is the pair of conditions section 3.3 asks for.

Setting. Full-batch descent, no weight decay, on the quadratic perceptron of appendix F: modular addition at $p = 97$, width 500, half the pairs held out, eight rates by two seeds, $30\,000$ steps, summarised in table 26. Two departures are deliberate. The log is written at *every* optimiser step, because the oscillation is the two-cycle of the unstable mode and its period is a few steps, while every other full-batch log here is strided at ten or more. The sharpness is measured along the run by power iteration on Hessian-vector products, taking the largest *algebraic* eigenvalue: the Hessian is indefinite early in training, when the most negative direction is the steeper, and plain power iteration then returns a negative sharpness. The routine agrees with an exact eigendecomposition on a 120-parameter model to 6×10^{-5} relative at the setting used.

Table 26: Whether undriven descent at the edge of stability oscillates, and whether it recurs. Full-batch descent by learning rate, both seeds. The stability ratio is $\eta\lambda_{\max}/2$ over the second half of the run; *rises* is the fraction of post-transition steps on which the loss increases; *return* is the nearest neighbour surviving a 150-sample exclusion divided by the distance the orbit travels during that exclusion, for which a two-frequency torus scores 0.008 and a monotone decay 1.007. In the middle band the sharpness self-limits at $2/\eta$: oscillation arrives in all eight runs there, recurrence in five of them.

η	stability ratio	rises	return	outcome
1×10^5	0.239, 0.238	0.000	1.007	monotone
3×10^5	0.758, 0.745	0.000	1.007, 1.007	monotone
1×10^6	0.969, 0.971	0.495, 0.490	0.104, 0.079	at the edge
1.5×10^6	0.974, 0.970	0.500, 0.500	1.012, 1.011	at the edge
2×10^6	0.977, 0.978	0.500, 0.500	0.120, 1.013	at the edge
2.5×10^6	—, 0.988	—, 0.500	—, 0.034	one seed diverges
2.8×10^6	0.992, —	0.503, —	0.022, —	one seed diverges
3×10^6	—	—	—	diverges

Oscillation is not recurrence. The oscillation is real and not round-off: on the one run whose step-by-step log is kept, the largest single-step rise is 3.1 times the loss itself, against a float32 relative epsilon of 1.2×10^{-7}. It is nonetheless a different property from recurrence. Eight runs reach the edge, and all eight oscillate at a rise fraction of one step in two; only five of them return near states they have left, while three trace a curve past the exclusion exactly as the monotone runs do. The return column of table 26 separates the two groups by a factor of ten with nothing between them, and places each group beside the control it belongs with.

Performance of the diagnostics. The admissible rectangle of section 3.4 rejects all eight runs at the edge, on the training loss and on the parameter norm alike, whether or not they recur. The estimate does not separate the two groups either: on the training loss it spans 3.8 to 8.6 in the five that return and 6.2 to 8.7 in the three that do not. The diagnostics are therefore silent on this regime rather than wrong about it, which is the weaker of the two failures section 3.4 allows for. The cause is visible in the inputs: the trend-crossing count tests non-monotonicity and not recurrence, so it passes every run at the edge, and ρ_{ident} stays between 1.5 and 2.6 throughout, far outside the admissible band in both groups. This is eight runs of one configuration on one task and we draw no rule from it.

Dependence on the logging stride. At the stride every other full-batch log here uses, most of the phenomenon is erased: over these eight runs the median trend-crossing count falls from 71 to 6 and the median rise fraction from 0.500 to 0.317. The collapse is not uniform, since the run with the most intermittent oscillation keeps a rise fraction of 0.46 under decimation, its cycle not being a clean two-cycle to alias. Whether a log looks like a transient is in part a property of how often it was written.

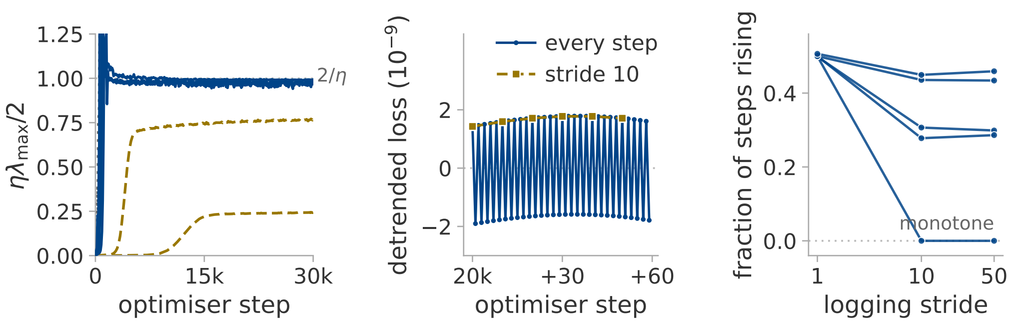

Figure 11: Whether full-batch descent reaches the edge of stability, and what the logging stride removes. (a) The stability ratio along training, seed 1 of each rate; the seeds agree to within the linewidth. Rates below the dotted line descend monotonically and are drawn as transients; those that pin against it are at the edge. (b) Sixty consecutive steps at $\eta = 2 \times 10^6$, against the same series at a stride of ten: the two-cycle is not blurred by that stride but removed by it. (c) The fraction of steps on which the loss rises, for each edge-of-stability rate, against the logging stride.

R The ceiling: embedding dimension against record length

Section 5.2 reports that the estimator saturates. Two explanations were available and the paper could not choose between them: the embedding condition $E > 2d$, which at $E_{\max} = 20$ allows ten components, and the finite-record bound $D \lesssim 2\log_{10} N$ (Eckmann & Ruelle, 1992), which at $N = 8000$ allows 7.8. They are separable because each is insensitive to the variable the other names. We vary $E_{\max} \in \{10, 14, 20, 28, 40, 56\}$ at $N = 8000$, and $N \in \{10^3 \ldots 6.4 \times 10^4\}$ at $E_{\max} = 20$, over $r \in \{2 \ldots 20\}$ and three seeds with everything else frozen; table 27 gives both. Each N is a *prefix* of one longer record, not a resampling of a fixed span, so the delay lag keeps its meaning in periods; all twenty-seven trajectories are bit-identical to the published ones over their common prefix, and rescoring the published twenty-direction configuration returns 15.44 against its published 15.1.

Table 27: Whether the ceiling follows the embedding condition or the finite-record bound. Two scans, three seeds per cell. *Tracking* is the largest r at which the median estimate stays within one component of the truth; *level* is the median estimate at $r = 20$; *slope* is $\mathrm{d}\hat{d}_{MG}/\mathrm{d}r$ over $r \geq 8$, which is 1 if the estimator still tracks the rank and 0 if it has stopped. The predictions each hypothesis makes for the tracking column are given beneath it.

$E_{\max}$ at $N = 8000$	10	14	20	28	40	56	
tracking	6.5	7.3	8.1	10.3	10.9	11.2	
$E_{\max}/2$ *predicts*	5	7	10	14	20	28	
level at $r = 20$	6.9	8.1	9.8	11.3	13.2	14.9	
slope	0.08	0.12	0.20	0.29	0.41	0.54	
N at $E_{\max} = 20$	10^3	$2{\cdot}10^3$	$4{\cdot}10^3$	$8{\cdot}10^3$	$1.6{\cdot}10^4$	$3.2{\cdot}10^4$	$6.4{\cdot}10^4$
tracking	6.7	8.3	8.5	8.1	8.9	9.5	9.5
$2\log_{10} N$ *predicts*	6.0	6.6	7.2	7.8	8.4	9.0	9.6
level at $r = 20$	9.0	9.3	9.4	9.8	9.9	10.2	10.5
slope	0.22	0.17	0.17	0.20	0.19	0.19	0.22

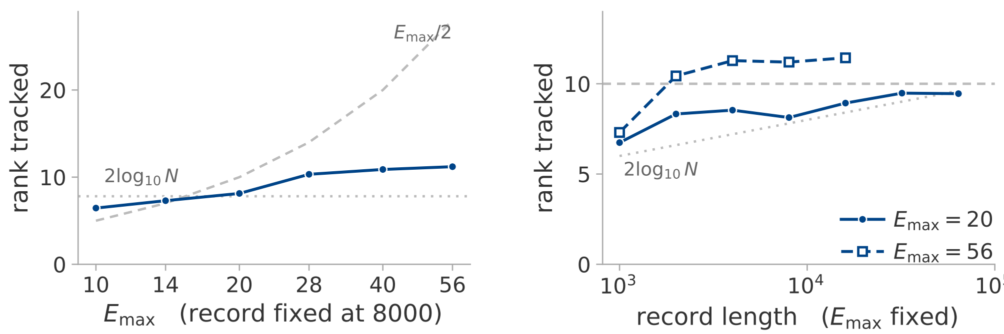

Figure 12: The rank the estimate tracks to, against the variable each hypothesis names and against what that hypothesis predicts. Every series is the constructed recurrent system, so all are drawn in one colour and the two predictions are reference lines. (a) Varying $E_{\max}$ at a fixed record: the measured ceiling rises far more slowly than $E_{\max}/2$, and the two cross near the $E_{\max} = 20$ this paper froze. (b) Varying the record at fixed $E_{\max}$. The dotted line is $2\log_{10} N$, the dashed line $E_{\max}/2$ for the filled series only. The open series decides it: at $E_{\max} = 56$ the estimate still tracks to 11.3 on a record of four thousand samples, whereas a finite-record bound allows 7.2.

Outcome of the two scans. The level at high rank belongs to the embedding: sixty-four times the record adds 1.4 components, whereas a factor of 5.6 in $E_{\max}$ adds 8.0. The tracking ceiling follows neither. It never passes 11.3 over the sixteen distinct $(E_{\max}, N)$ cells, the two scans above plus a third that repeats the record scan at $E_{\max} = 56$, and the cell that attains it is $E_{\max} = 56$ at $N = 4000$, half the record the rest of the study uses, where $2\log_{10} N$ allows 7.2. A record-length bound cannot produce that, so Eckmann & Ruelle (1992) is not the explanation here. Nor is $E > 2d$ in the form we used it: the ceiling rises by 0.10 components per unit of $E_{\max}$, not by a half, so $E_{\max}/2$ happens to be right at $E_{\max} = 20$ and is wrong at both ends — which is Sauer & Yorke's point in the other direction, that how many delay coordinates one needs does not follow from the embedding theorem alone (Sauer & Yorke, 1993).

Fitting the level at $r \geq 16$ separates the candidates (table 28). Neither hypothesis fits; their pointwise minimum fits no better, so they do not jointly bind either; and a form logarithmic in both variables fits an order of magnitude inside the spread of the data itself. Both dependences are therefore logarithmic: 2.6 components per doubling of $E_{\max}$, 0.9 per decade of record.

Table 28: Whether either candidate explanation of the ceiling accounts for the measured level. Candidate forms fitted at $r \geq 16$ over the sixteen $(E_{\max}, N)$ cells; the spread of the fitted quantity itself is 2.00 components, so a form with a larger error than that describes nothing.

form	root-mean-square error
$E_{\max}/2$, the embedding condition	8.73
$2\log_{10} N$, the finite-record bound	3.68
the pointwise minimum of the two	3.70
$8.61\log_{10} E_{\max} + 0.90\log_{10} N - 5.31$	0.19

Two qualifications. First, $E_{\max}$, τ and the delay span $(E_{\max} - 1)\tau$ are algebraically dependent. A third scan holding the span at 76 while τ falls from 8 to 1 finds that $E_{\max}$ then adds nothing (regression slope 0.000). The operative variable appears to be the temporal extent of the delay window, which is not what $E > 2d$ constrains either; that scan is defective at both endpoints, so it does not settle the converse.

Second and more seriously, *the linear null reproduces the whole dependence*. $\mathrm{PR}_{\mathrm{delay}}$, a spectral statistic of the delay covariance with no geometry in it and capped only by E, rises from 5.33 to 13.53 across the $E_{\max}$ scan where MG rises from 6.94 to 13.55, and is flat to two decimals across the record scan. It is fitted by $11.09\log_{10} E_{\max}$ with a root-mean-square error of 0.09, the tightest fit anywhere in this appendix. Most of what we have been calling a ceiling is therefore how many components the delay window resolves *linearly*, which is a spectral account of the saturation and not a geometric one.

The unexplained tracking limit. We cannot say why the tracking ceiling lies near eleven components. It is not $E_{\max}/2$, since it is 11.2 at $E_{\max} = 56$; not $2\log_{10} N$, since it is 11.3 at $N = 4000$; and not the measured effective rank, which equals r to 0.21 at every $N \geq 2000$. No form we fitted accounts for it, and nothing in the setup explains it: the Theiler axis is inert (a fixed-76 scan reproduces the frozen setting to two decimals at every cell), and none of the 270 cells is marked degenerate. This is a real limit of the estimator, which we can bound but not explain.

Section 3.4's own ratio does see it: ρ_{ident} at $r = 20$ is 1.12 to 1.36 at every record length, a longer record never making an unidentifiable estimate identifiable, and falls from 1.31 to 1.03 as $E_{\max}$ grows, crossing the band at about the rank where tracking stops.

S Two controls separating the estimator from roughness

Roughness, $\mathrm{std}(\Delta x)/\mathrm{std}(x)$, follows the estimate closely across the constructed systems of section 5, and it registers the grokking transition more sharply than the estimate does (section 7.3). Neither observation can be interpreted until a prior question is settled: whether the two statistics are separable at all. The two controls below settle it in opposite directions, and each was given its pass criterion before it was run.

The count changes and the roughness does not. Both arms are a sum of four sinusoids. In the first the frequencies are rationally independent, so the orbit closes on a four-torus and $d_{\mathrm{act}} = 4$. In the second they are harmonics of one irrational base, so a single phase fixes all four hands, the orbit is a circle, and $d_{\mathrm{act}} = 1$: four components in the signal, one degree of freedom behind them. Bisection then sets the one-clock base frequency so that the two arms have the *same* roughness, to a part in 10^{15}, which leaves a roughness statistic nothing to separate them by. A scheduled run passes $4 \to 1 \to 4$ through ramped envelopes, and every window that a ramp reaches into is left unscored.

The roughness changes and the count does not. Five monotone observers $g(x) = x + ax^p$ are applied to one four-phase system. Each is invertible, so it may bend the recorded set but cannot add or remove a coordinate. Each also changes the local scale $g'(x)$, and with it the spacing of consecutive samples.

Table 29: Both controls, median over eight seeds. Above the rule, two arms matched in roughness and differing in the number of independent phases. Below it, one four-phase system under five invertible maps. $\mathrm{PR}_{\mathrm{delay}}$ is the linear participation ratio of the delay covariance, included because it follows the roughness rather than the active dimension in both halves.

	d_{act}	estimate	roughness	$\mathrm{PR}_{\mathrm{delay}}$
four independent clocks	4	3.92	0.091	2.20
one clock, four hands	1	1.11	0.091	4.10
identity	4	3.92	0.091	2.20
$x + 0.25x^3$	4	3.96	0.098	2.32
$x + x^3$	4	3.96	0.110	2.58
$x + x^5$	4	4.01	0.146	3.61
$x + 0.1x^7$	4	4.02	0.166	4.28

Both pre-specified criteria hold: the estimate follows the count while roughness moves by under a tenth of a per cent, and roughness spans $83\,\%$ while the median estimate stays within 0.08 of four.

What this does not settle. The two controls separate the statistics on a constructed system whose phases are known. They say nothing about a training log, where the surrogate control of section 7.3 carries the claim instead. Roughness measures how fast the recorded value moves; the estimate attempts to count how many independent phases move it.

Two results go against the estimator. The first is the inversion in table 29: the linear participation ratio reads higher on the circle than on the four-torus, so anyone treating it as a mode count would order the two backwards. The second concerns the circle itself, on which the estimate turns out not to be geometric. Every admissible neighbour there is a sample from another pass, and the local speed of the curve multiplies all of a point's neighbour distances alike, so speed cancels from the log-ratios of equation (3). What remains depends on the arithmetic of the frequency. Two of the eight seeds in the first run of this control drew a base frequency whose period has a large early partial quotient, so its multiples cluster into near-duplicates and the estimate falls to 0.90 rather than the 1.11 the other seven return. The value is reproducible from the fractional parts of multiples of that frequency alone, with no signal and no neighbour search. On a closed orbit the estimator is therefore sensitive to a number-theoretic accident of the period, which should be known before a value near one is read as confirming a circle.

T Compute cost and code availability

Compute cost. Everything here runs on one consumer GPU and one desktop CPU, in hours rather than days. The largest GPU item is the edge-of-stability campaign of appendix Q: sixteen runs of $30\,000$ full-batch steps, logged at every step and with the sharpness measured three hundred times per run, in 1770 seconds total on a single T4. The analysis is the more expensive half and is all CPU, a few core-hours per experiment, the ceiling scan of appendix R the largest single item. Nothing in this paper needs a cluster, which is deliberate: the estimator is meant to be inexpensive enough to run on a log one already has.

Cost of the trajectory sketch. The claim of section 7.2 that measuring the trajectory directly is affordable should be given a number. In storage it is exact arithmetic: at each logged step the sketch keeps 1024 coordinates in each of two spaces under two hash families, 4096 floats, against the full parameter vectors of appendix O: a factor of 36 for the perceptron and 55 for the transformer. Over the 151 logged steps of a 1500-step run that is 2.5 against 88 megabytes.

In time the overhead is below what we can measure. Training the same configuration with and without the observer attached, alternating the order so that warm-up cannot favour either, and checking that the two agree on every logged loss to the last bit, gives 41.75 seconds bare against 40.44 probed — a nominal overhead of $-3.1\,\%$. A negative overhead is not a result but a bound: at the logging stride the transformer runs use, the cost of the sketch is under the run-to-run variation of the machine, which here is a few per cent. `sketch_cost.py` records the measurement.

Code availability. All code, configurations and result files behind every number and figure in this paper will be released publicly on acceptance: the estimator and its frozen configurations, the six constructed systems, the trajectory sketch, the training scripts for both grokking settings, and the analysis scripts named throughout these appendices. The figures regenerate from the committed result files with a single command.